Kondo and RKKY interactions in bilayer graphene

Aravindh Swaminathan Shankar

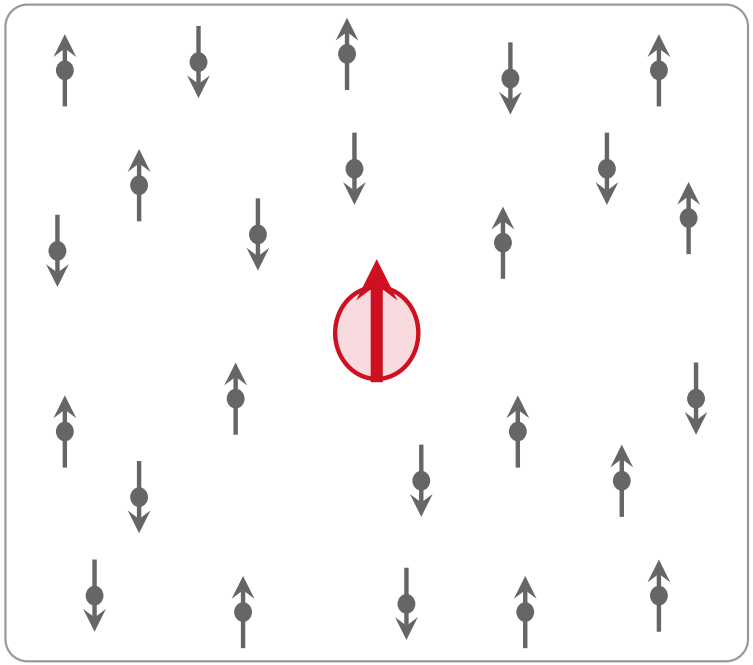
ashankar@ictp.it

with D. O. Oriekhov, L. Fritz, A. K. Mitchell

Abdus Salam ICTP, Trieste, Italy

NM4QM 2026

A magnetic impurity in a metal is screened away below T_K: the Kondo effect

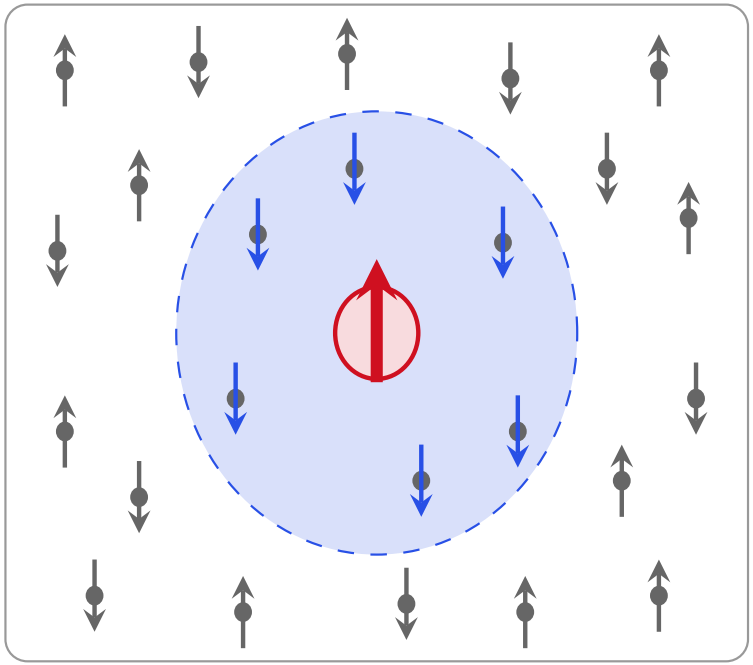


local moment

$$H = H_{\text{host}} + J\,\vec{S}\cdot\vec{s}(0)$$

Impurity spin coupled to the conduction electrons

A magnetic impurity in a metal is screened away below T_K: the Kondo effect



Kondo singlet

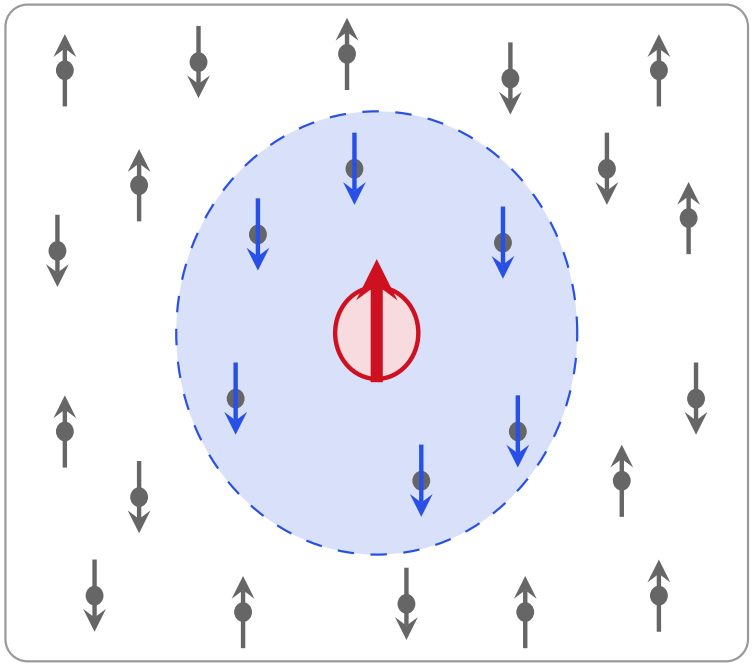
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Low T: the electrons rearrange and screen the moment

Strong coupling: beyond perturbation theory

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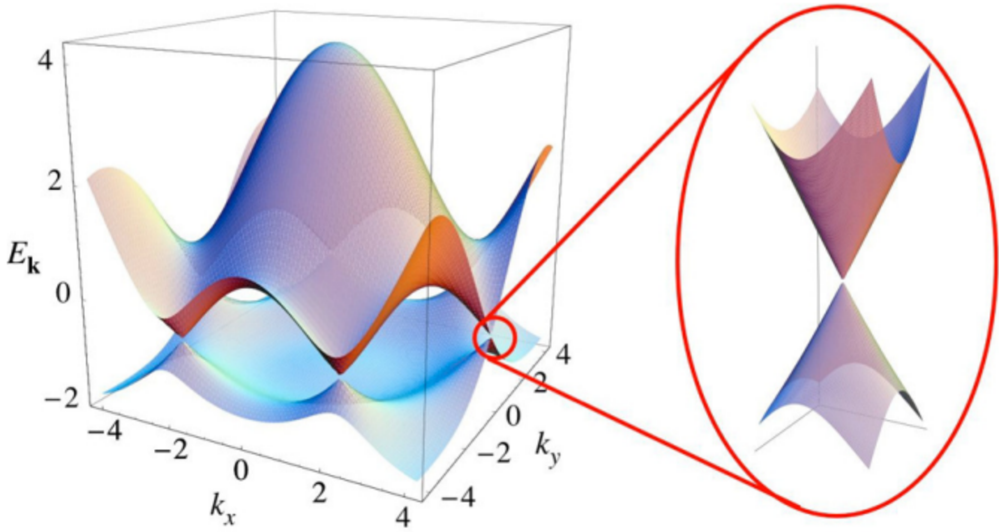
Strong coupling: beyond perturbation theory

$$T_K = D\,e^{-1/\rho_0 J}$$

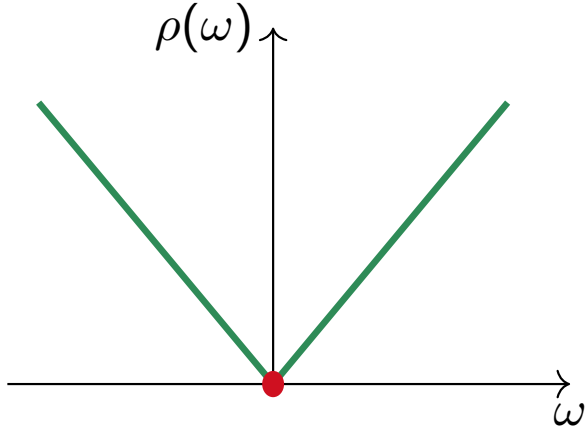
host density of states at the Fermi level

The host density of states decides if, and when, the moment is screened

No Kondo effect in graphene at charge neutrality



Castro Neto et al., Rev. Mod. Phys. 81, 109 (2009)



$\rho(E_F) = 0$

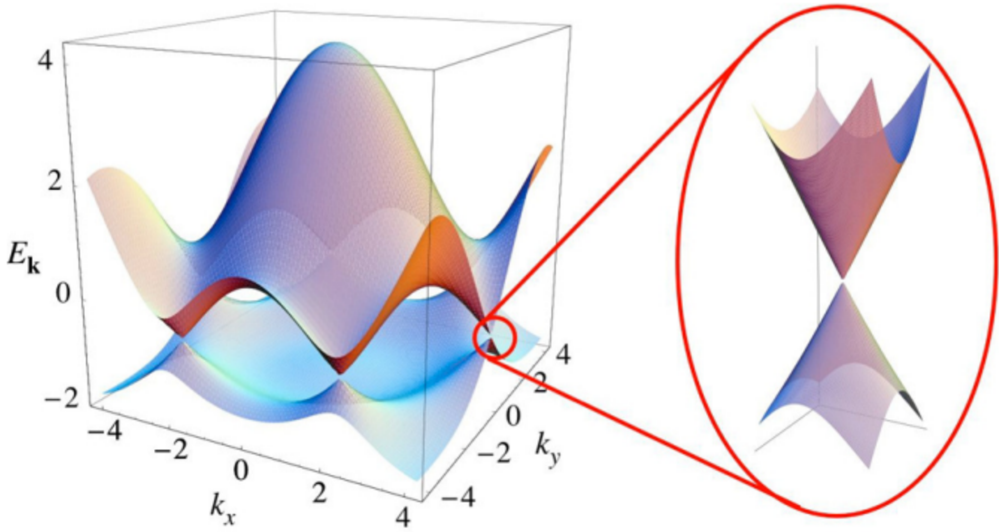
$$T_K = D\, e^{-1/\rho_0 J}$$

$\rho_0 \to 0 \;\Rightarrow\; T_K \to 0$

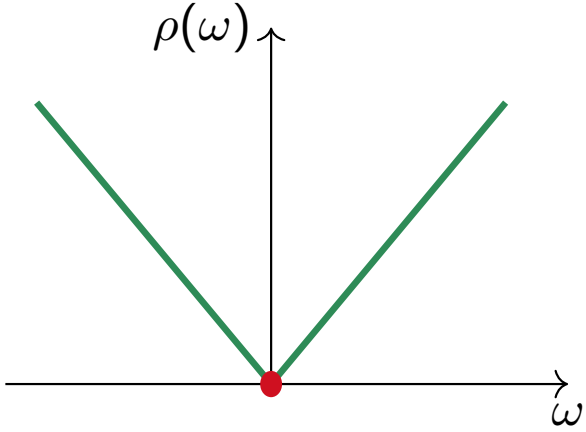
the moment stays free

What happens in other graphene-like materials?

No Kondo effect in graphene at charge neutrality



Castro Neto et al., Rev. Mod. Phys. 81, 109 (2009)



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Experiments in graphene

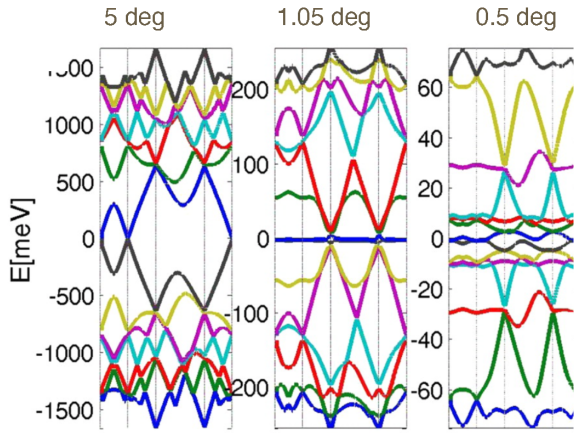
H adatoms: González-Herrero et al., Science 352, 437 (2016)
Vacancies: Jiang, . . . , Andrei, Nat. Commun. 9, 2349 (2018)

What happens in other graphene-like materials?

Model: Anderson impurity on twisted bilayer graphene

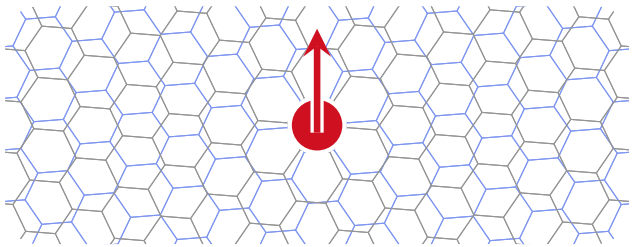
Host Bistritzer–MacDonald continuum model

$$H(\mathbf{k}) = \begin{pmatrix} h^K_{\theta/2}(\mathbf{k}) & wT_1 & wT_2 & wT_3 \\ wT_1^\dagger & h^K_{-\theta/2}(\mathbf{k}-\mathbf{q}_1) & 0 & 0 \\ wT_2^\dagger & 0 & h^K_{-\theta/2}(\mathbf{k}-\mathbf{q}_2) & 0 \\ wT_3^\dagger & 0 & 0 & h^K_{-\theta/2}(\mathbf{k}-\mathbf{q}_3) \end{pmatrix}$$



Bistritzer and MacDonald, PNAS 108, 12233 (2011)

Impurity Anderson model



$$H_{\text{imp}} = \epsilon_d \sum_\sigma n_{d\sigma} + U\, n_{d\uparrow} n_{d\downarrow} + V \sum_\sigma \left(d_\sigma^\dagger c_{0\sigma} + \text{h.c.} \right)$$

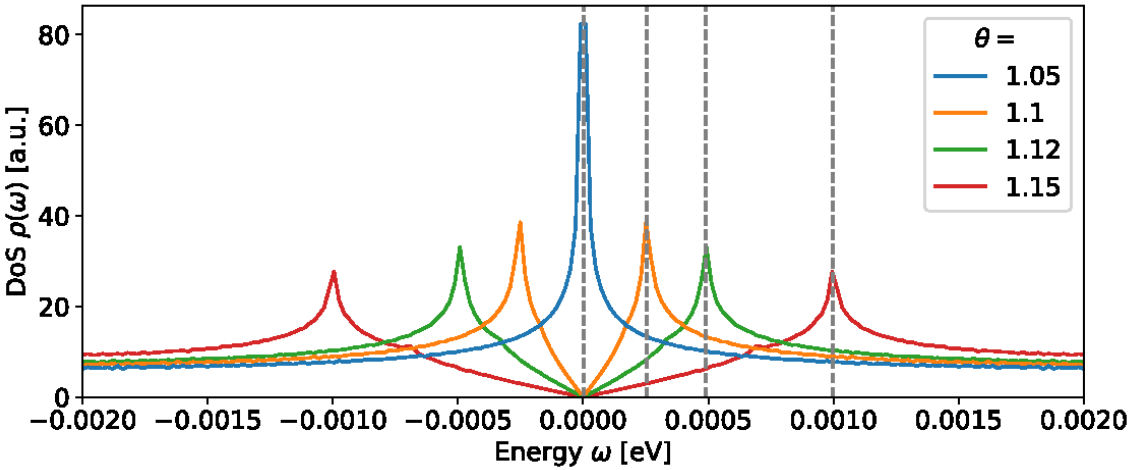
$|0\rangle$ $|\uparrow\rangle$ $|\downarrow\rangle$ $|\uparrow\downarrow\rangle$

4 states; large U leaves $\uparrow, \downarrow$:
a spin-$\frac{1}{2}$ moment, $J \sim V^2/U$

Impurities in TBG: Shankar, Oriekhov, Mitchell, Fritz, PRB 107, 245102 (2023);

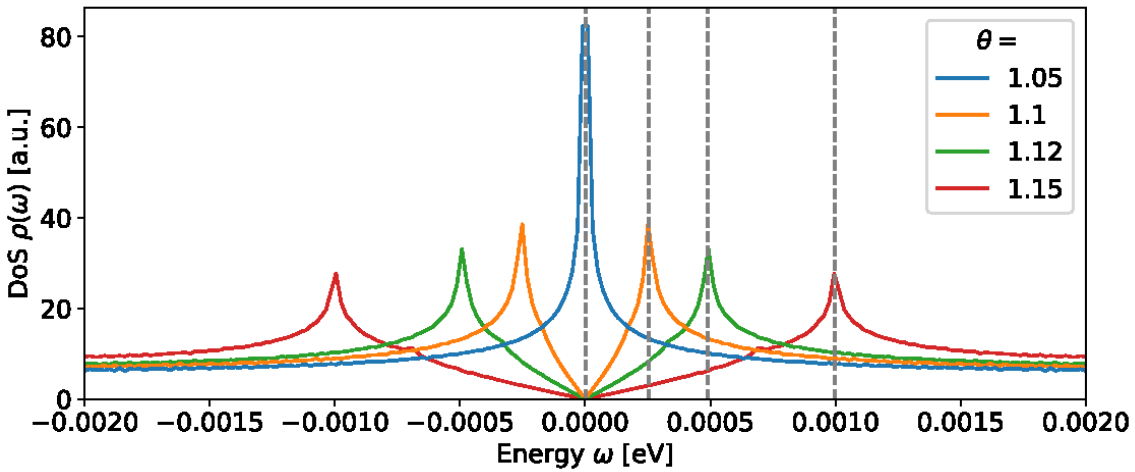
Chang, Yi, Wu, Kugler, Andrei, Vanderbilt, Kotliar, Pixley, PRL 133, 126503 (2024)

van Hove singularities in the flat band



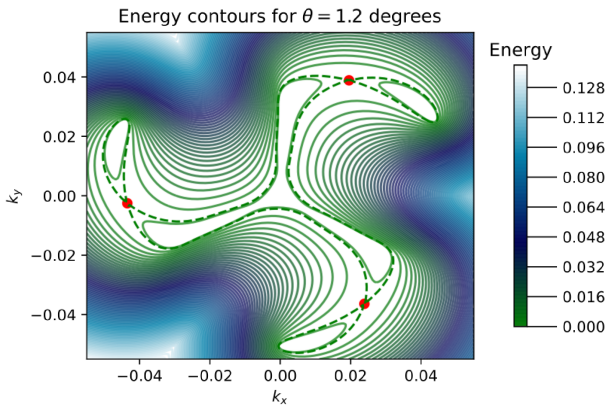
Dirac cone at $\omega = 0$: $\rho \sim |\omega|$

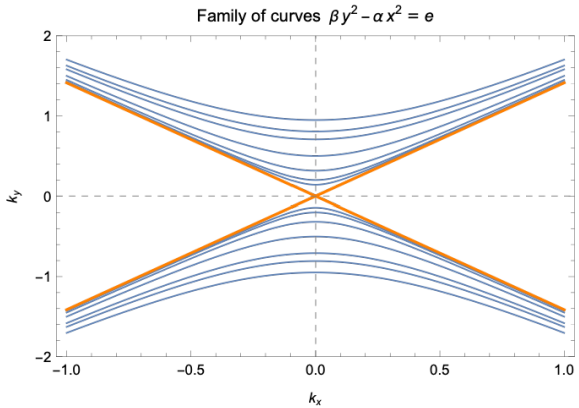
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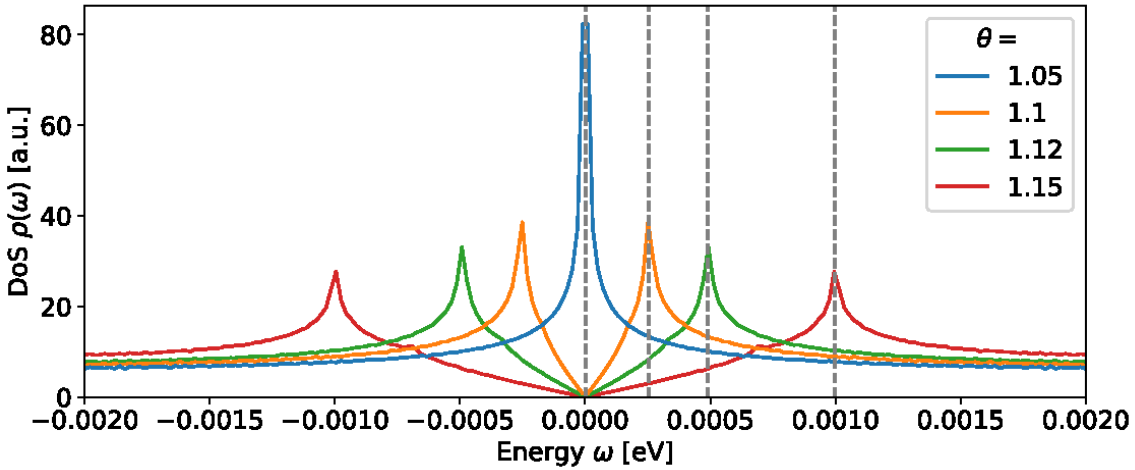
van Hove peaks at $\pm E_v$





ordinary saddle $\quad E = E_v + \alpha p_x^2 - \beta p_y^2 \implies \rho(E) \sim \log \left| \frac{\Lambda}{E - E_v} \right|$

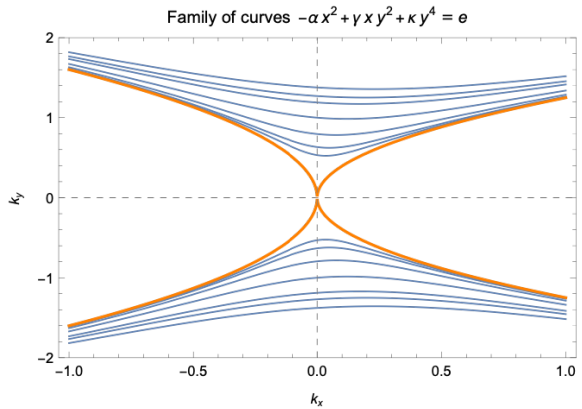
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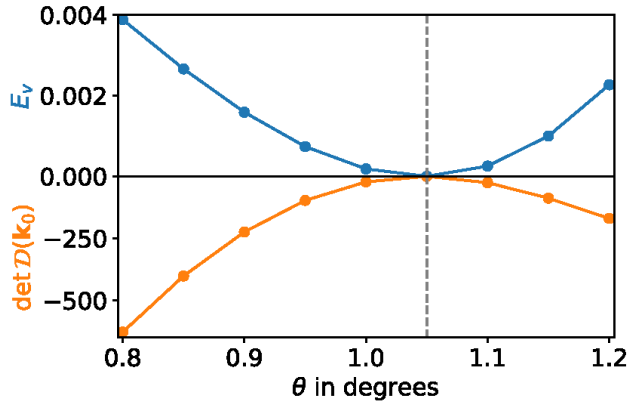


Dirac cone at $\omega = 0$: $\rho \sim |\omega|$

van Hove peaks at $\pm E_v$

1.05°: the peaks merge





higher-order saddle $\quad E = \alpha p_x^2 + \gamma p_x p_y^2 + \kappa p_y^4 \implies \rho(E) \sim |E|^{-1/4}$

Yuan, Isobe, Fu, Nat. Commun. 10, 5769 (2019)

At 1.05° the two vHs merge into a higher-order vHs

A van Hove peak boosts T_K

	Density of states $\rho(\omega)$	T_K	$S_{\rm imp}(T \to 0)$
Metal	ρ_0 : constant	$D\, e^{-1/\rho_0 J}$	0
Dirac cone	$\rho_0 \lvert\omega\rvert$: linear pseudogap	$T_K = 0$	$\ln 2$
Standard van Hove	$\rho_0 \left[1 + a \ln(D/\lvert\omega\rvert)\right]$: log divergence	$D\, e^{\frac{1}{a} - \frac{1}{a}\sqrt{1+\frac{2a}{\rho_0 J}}}$	0
Higher-order van Hove	$\rho_0 \lvert\omega\rvert^{-\alpha}$: power-law divergence	$D \left(1 + \frac{\alpha D^\alpha}{\rho_0 J}\right)^{-1/\alpha}$	$-2\alpha \ln 2$

Mitchell, Vojta, Bulla, Fritz, PRB 88, 195119 (2013)

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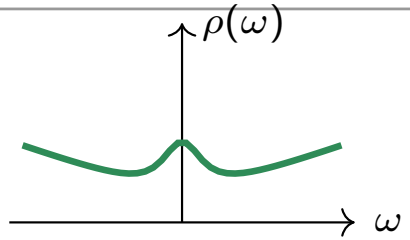
Mitchell, Vojta, Bulla, Fritz, PRB 88, 195119 (2013)

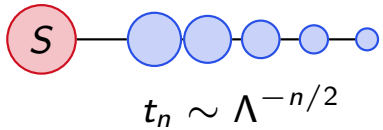
Logarithmic vHs: T_K enhanced over the metal

Higher-order vHs: T_K grows as a power law in J

In TBG the cone and the vHs sit within a few meV of each other: which one wins?

NRG settles it non-perturbatively: read the answer off the impurity entropy

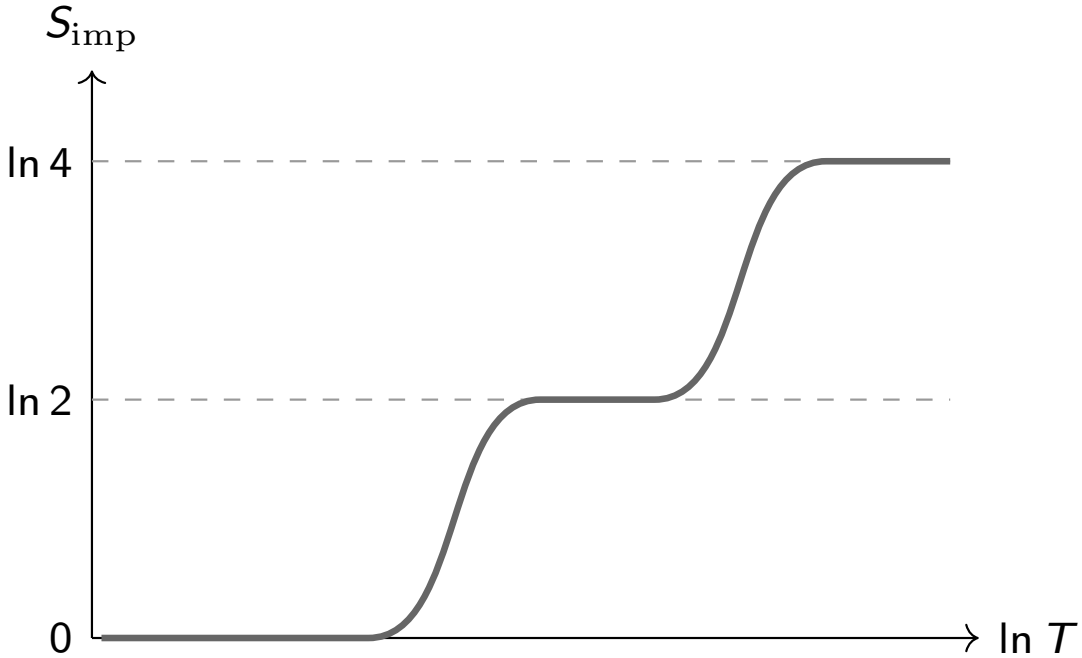




3. $S_{\rm imp}(T)$

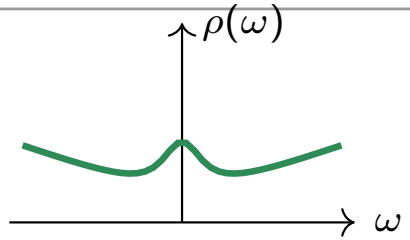
1. host DoS $\rho(\omega)$

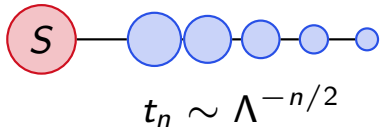
2. Wilson chain, solved iteratively



A plateau at ln **N**: **N states** available to the impurity

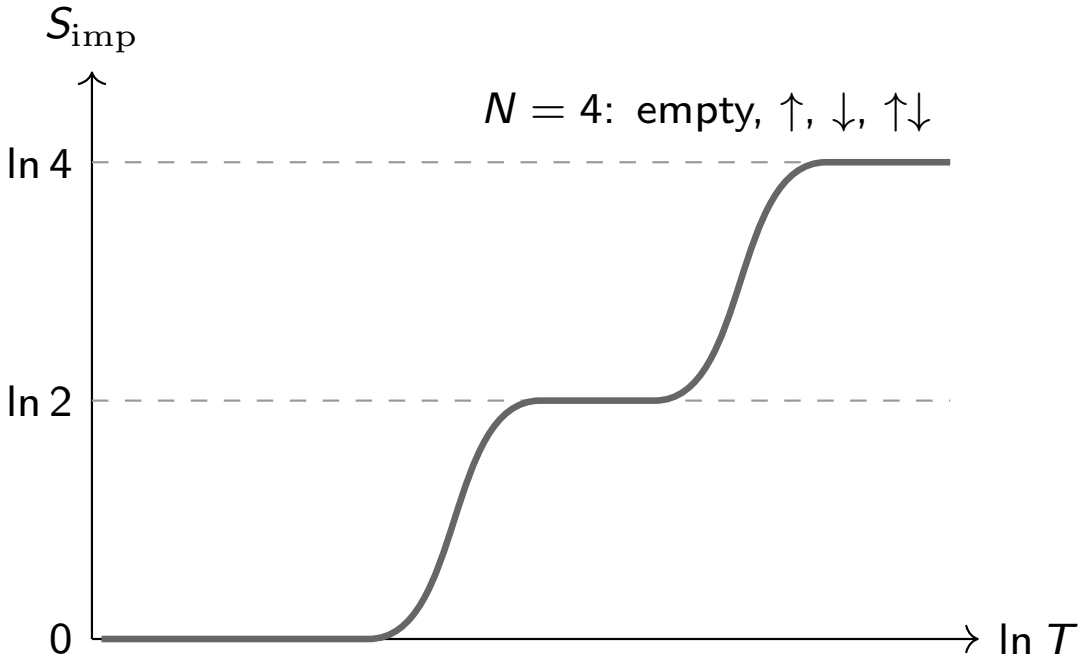
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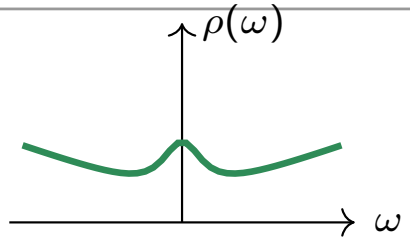
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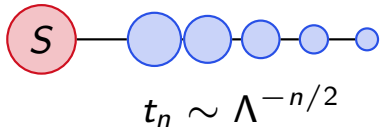


A plateau at ln **N**: **N states** available to the impurity

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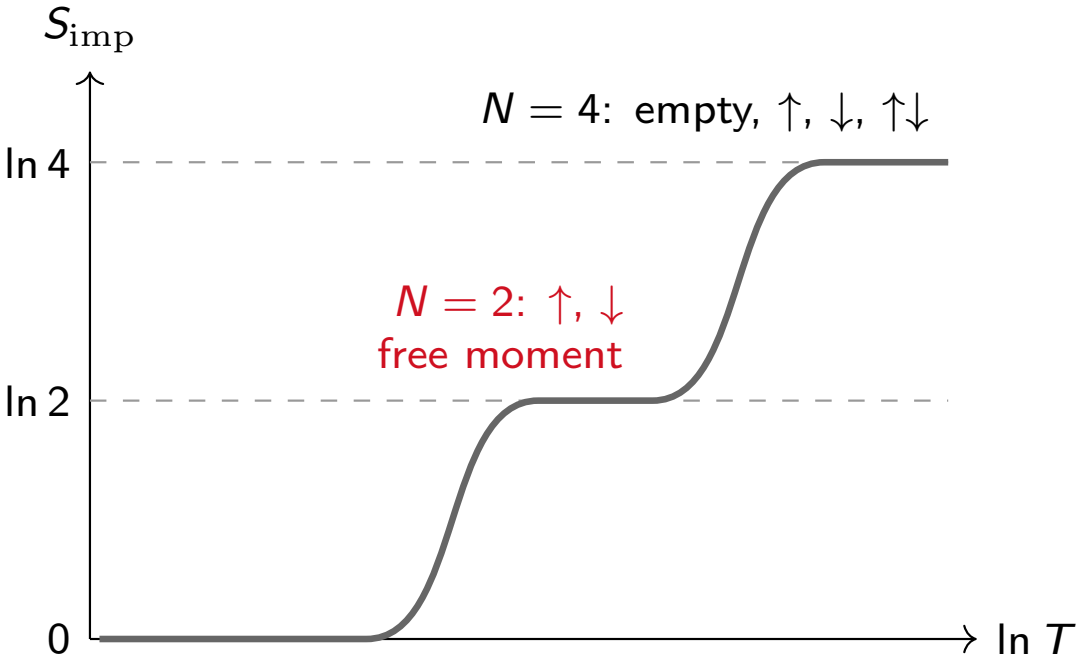


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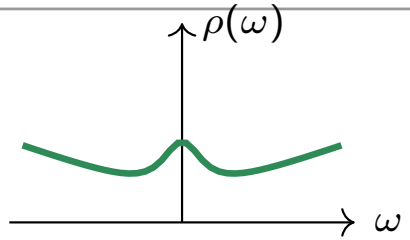
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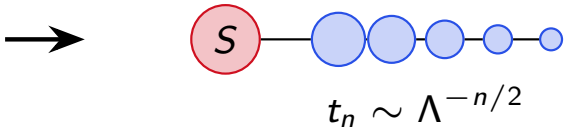


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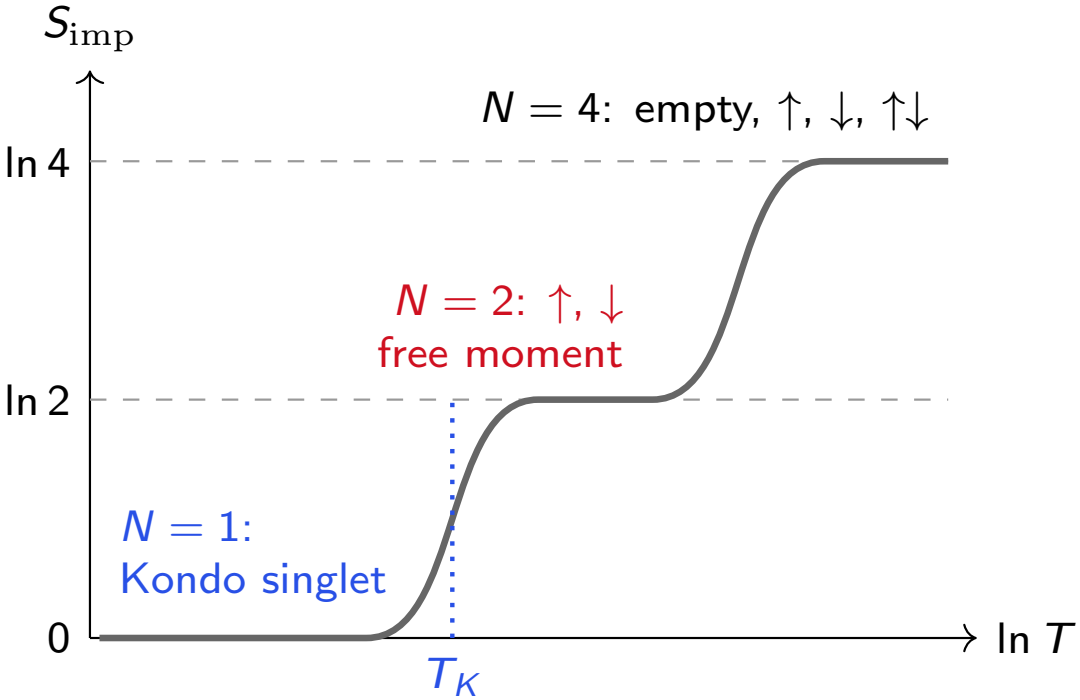


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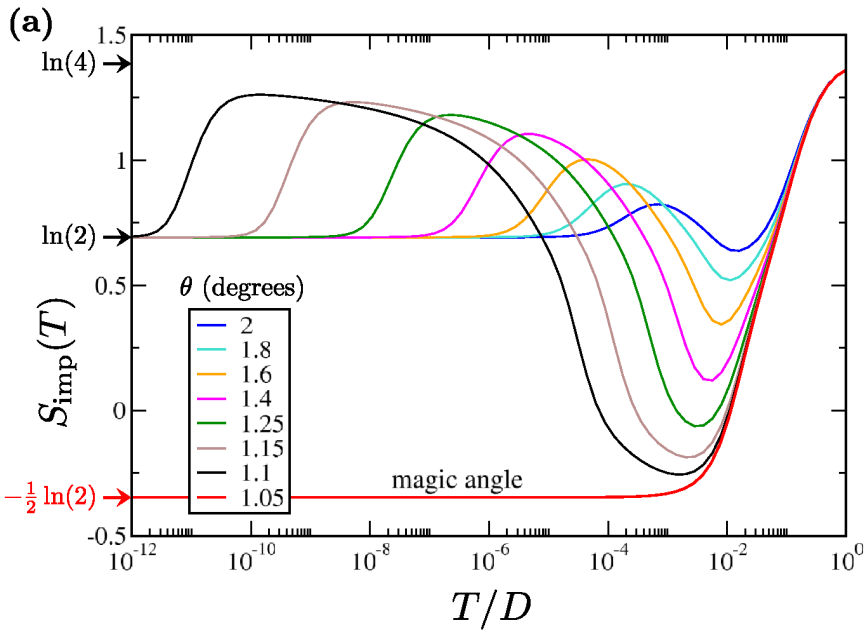
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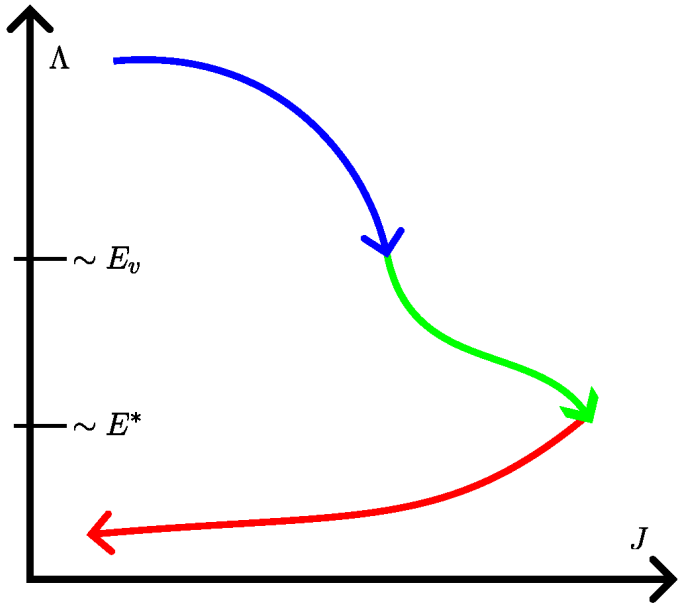
A step between plateaus: a screening scale, here T_K

$S_{\mathrm{imp}} = S_{\mathrm{total}} - S_{\mathrm{host}}$, so it can be negative

This curve is the one diagnostic used for the rest of the talk

Kondo revival at the magic angle



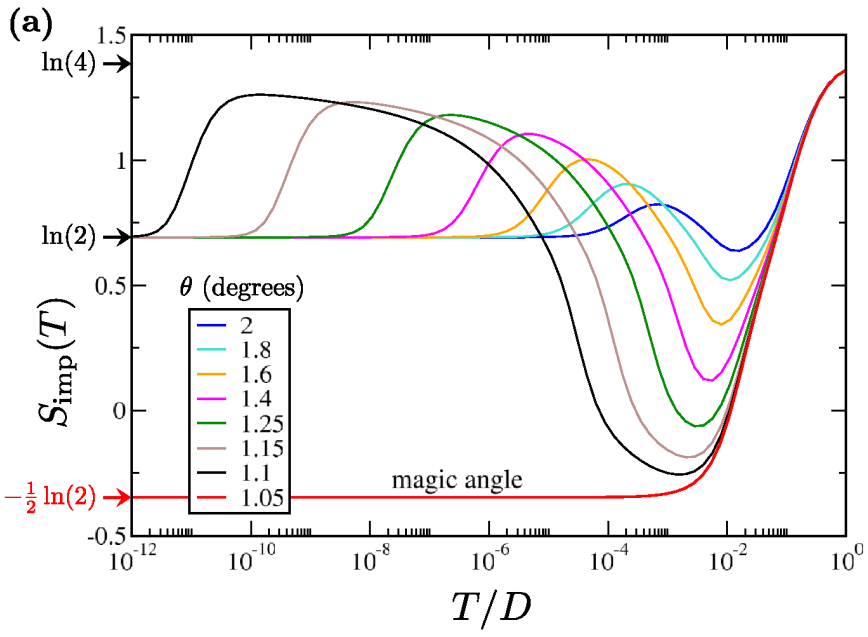


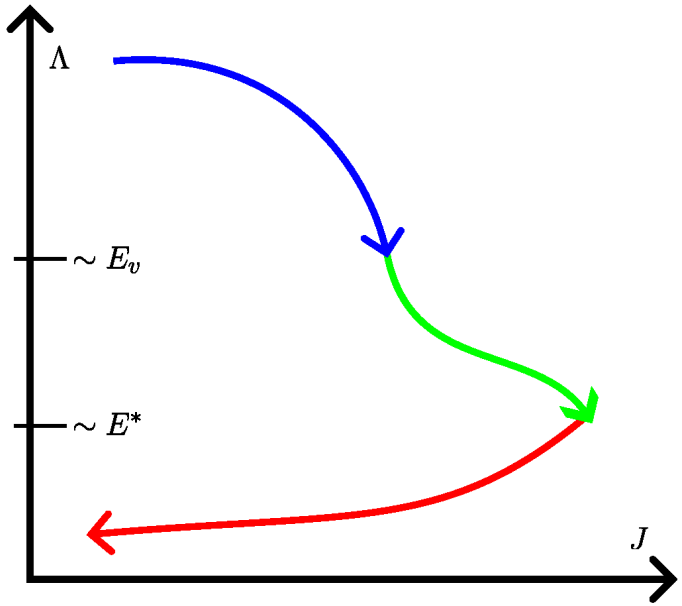
Shankar, Oriekhov, Mitchell, Fritz, PRB 107, 245102 (2023)

See also: Chang, Yi, Wu, Kugler, Andrei, Vanderbilt, Kotliar, Pixley, PRL 133, 126503 (2024)

Screening begins at E_v; below $E^* \sim |E_v|^3$ the cone wins: moment unscreened

Kondo revival at the magic angle





Shankar, Oriekhov, Mitchell, Fritz, PRB 107, 245102 (2023)

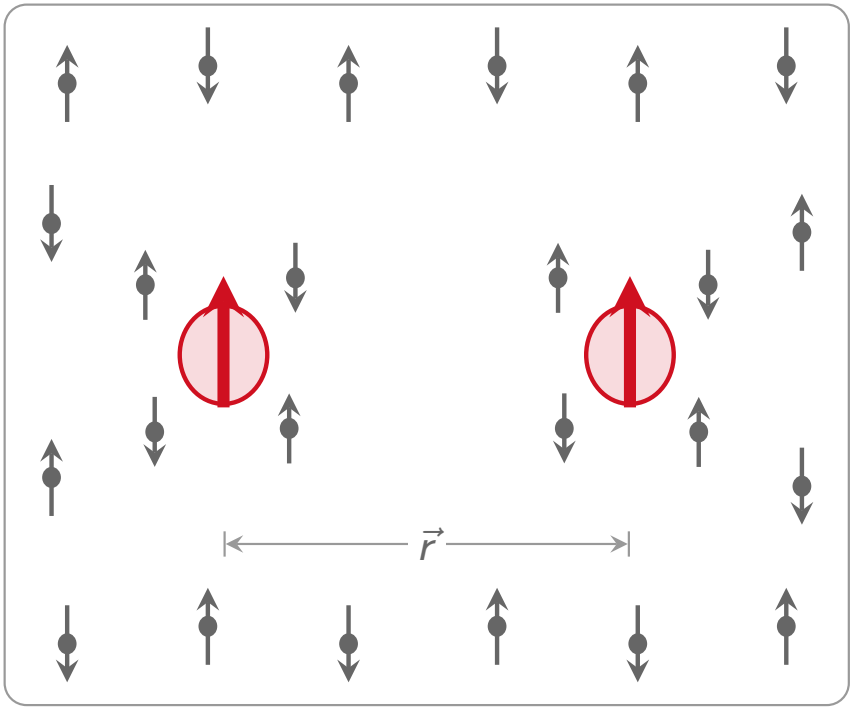
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Screening begins at E_v; below $E^* \sim |E_v|^3$ the cone wins: moment unscreened

Magic angle: E_v, $E^* \to 0$, the cone never wins: moment screened

One impurity is only the dilute limit. What happens when two sit close together?

Two impurities: the RKKY interaction

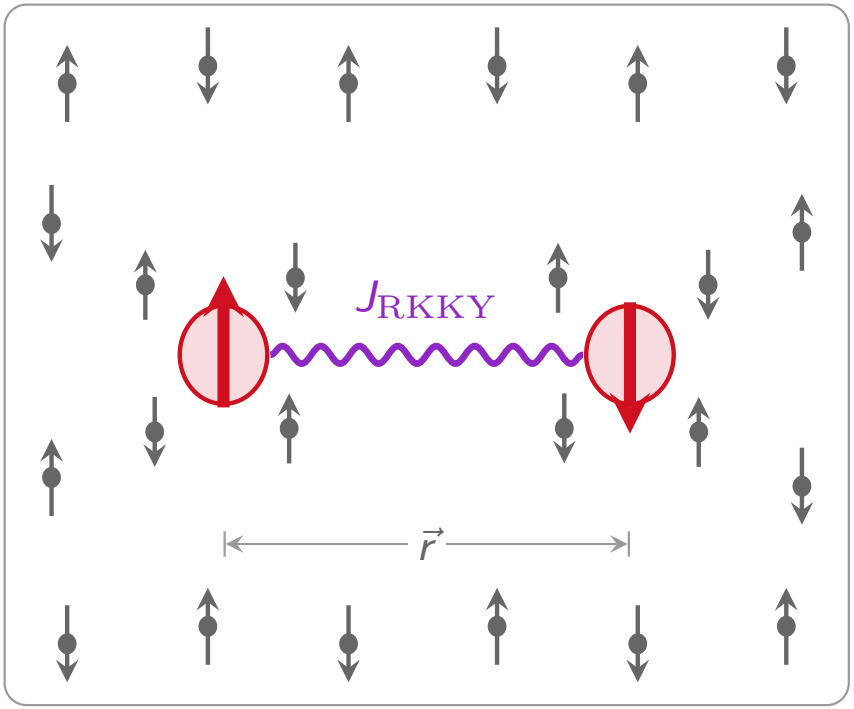


two local moments

$$H = H_{\text{host}} + J \sum_{i=1,2} \vec{S}_i \cdot \vec{s}(\vec{r}_i)$$

Each spin couples to the electrons at its own site

Two impurities: the RKKY interaction



the moments lock together

Jayaprakash, Krishna-murthy, Wilkins, PRL 47, 737 (1981)
Jones, Varma, Wilkins, PRL 61, 125 (1988)
STM experiment: Spinelli et al., Nat. Commun. 6, 10046 (2015)

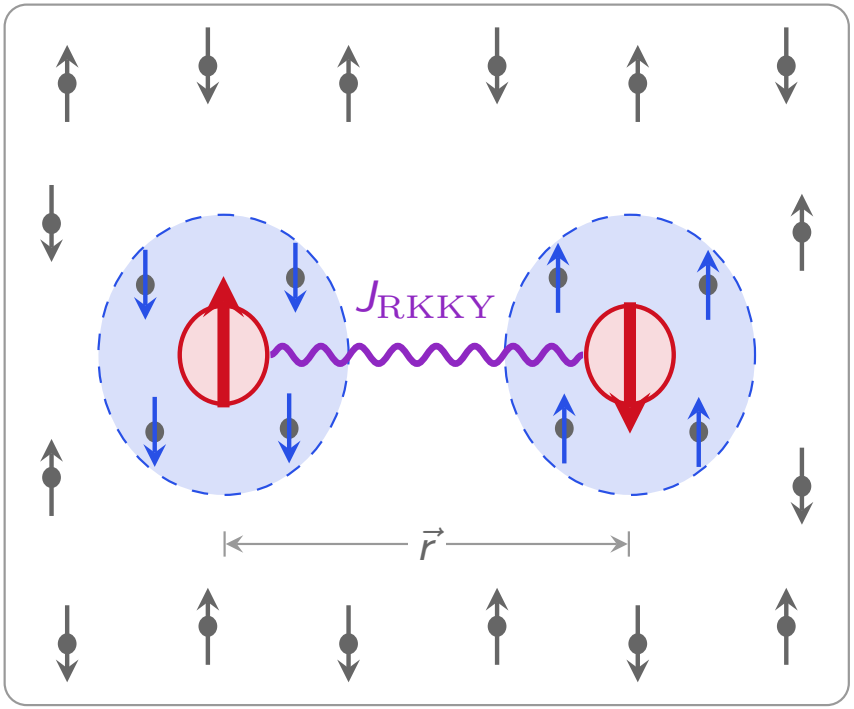
$$H = H_{\mathrm{host}} + J \sum_{i=1,2} \vec{S}_i \cdot \vec{s}(\vec{r}_i)$$

Each spin couples to the electrons at its own site

The electrons carry an effective exchange $J_{\mathrm{RKKY}}\, \vec{S}_1 \cdot \vec{S}_2$

$$J_{\mathrm{RKKY}} \simeq 2J^2 \int_{\Omega_1 < E_F < \Omega_2} \frac{d\Omega_1 d\Omega_2}{\pi^2} \frac{\rho(-\vec{r}, \Omega_1)\, \rho(\vec{r}, \Omega_2)}{\Omega_1 - \Omega_2}$$

Two impurities: the RKKY interaction



or each one Kondo-screened?

Jayaprakash, Krishna-murthy, Wilkins, PRL 47, 737 (1981)
Jones, Varma, Wilkins, PRL 61, 125 (1988)
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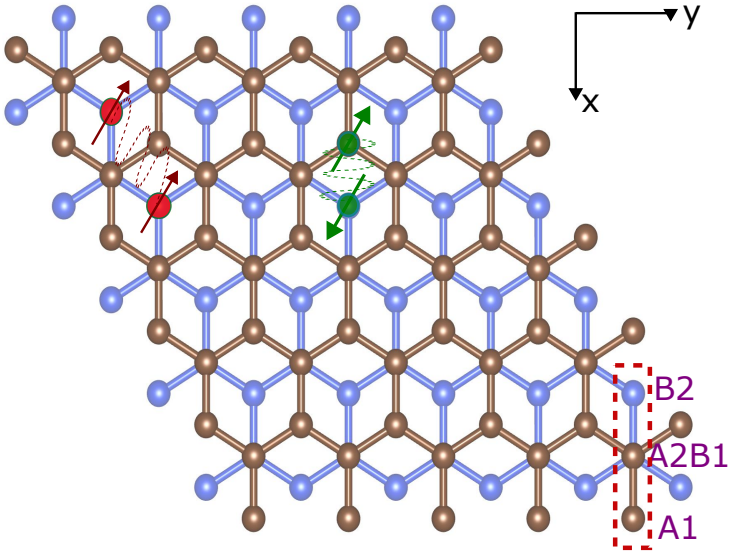
$$J_{\mathrm{RKKY}} \simeq 2J^2 \int_{\Omega_1 < E_F < \Omega_2} \frac{d\Omega_1 d\Omega_2}{\pi^2} \frac{\rho(-\vec{r}, \Omega_1)\, \rho(\vec{r}, \Omega_2)}{\Omega_1 - \Omega_2}$$

Needed: $\rho(\vec{r}, \omega) = -\,\mathrm{Im}\, G(\vec{r}, \omega)$

non-local Green's function between the two sites, at all energies

RKKY $\propto J^2$, Kondo $\sim e^{-1/\rho_0 J}$: which one wins?

Bernal bilayer graphene, no twist



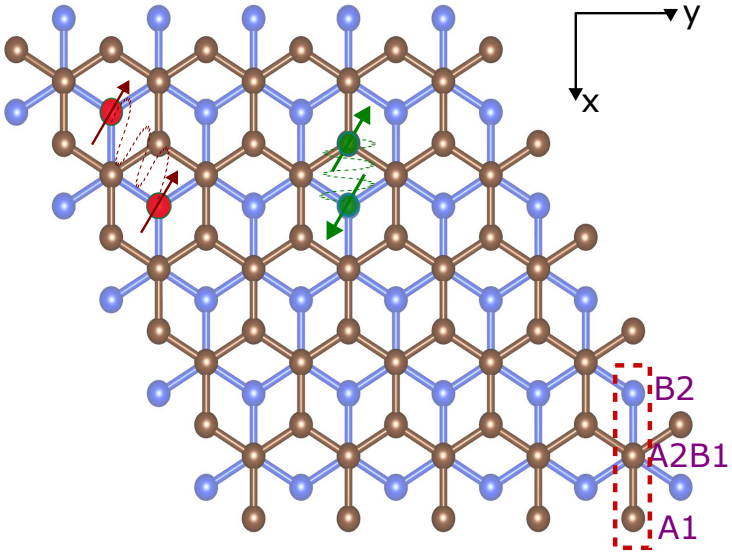
4-band tight binding, fitted to infrared spectra

parameter	value (eV)	what it does
γ_0	3.16	in-plane A–B hopping
γ_1	0.381	vertical B1–A2 (stacked sites)
γ_3	0.38	skew A1–B2: trigonal warping
γ_4	0.14	skew A1–A2, B1–B2: e–h asym.
Δ'	0.022	stacked-site energy: e–h asym.

Model: McCann, Koshino, Rep. Prog. Phys. 76, 056503 (2013)
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RKKY in bilayer graphene:
Jiang et al., J. Phys.: Condens. Matter 24, 206003 (2012)
Parhizgar, Sherafati, Asgari, Satpathy, PRB 87, 165429 (2013)
Klier, Shallcross, Pankratov, PRB 90, 245118 (2014)

Bernal bilayer graphene, no twist

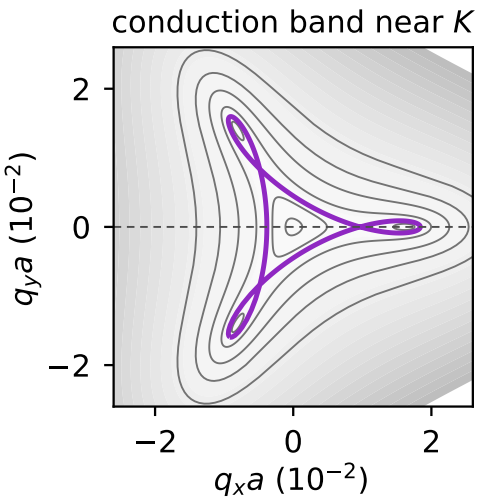


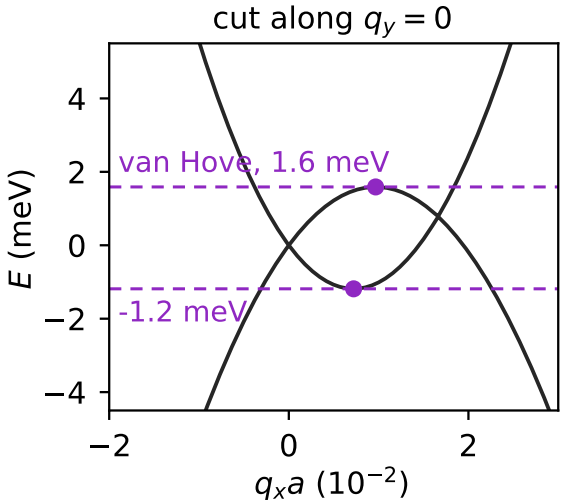
4-band tight binding, fitted to infrared spectra

parameter	value (eV)	what it does
γ_0	3.16	in-plane A–B hopping
γ_1	0.381	vertical B1–A2 (stacked sites)
γ_3	0.38	skew A1–B2: trigonal warping
γ_4	0.14	skew A1–A2, B1–B2: e–h asym.
Δ'	0.022	stacked-site energy: e–h asym.

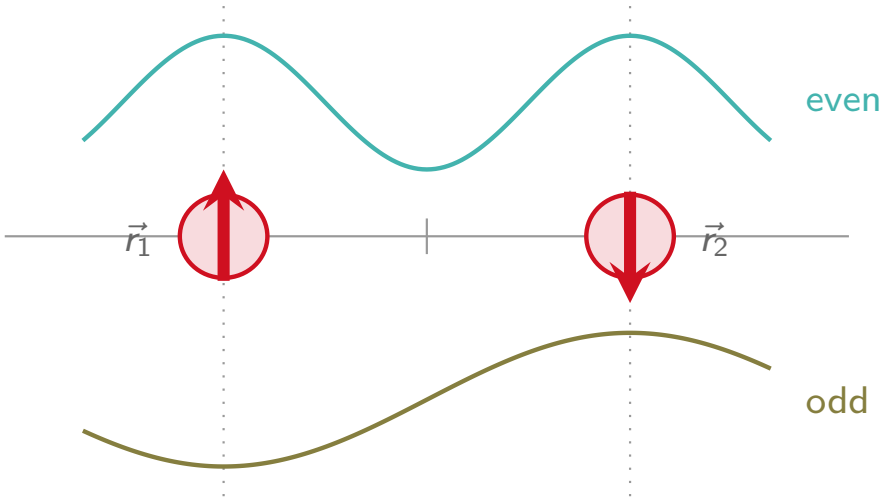
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NRG for two impurities: even and odd channels



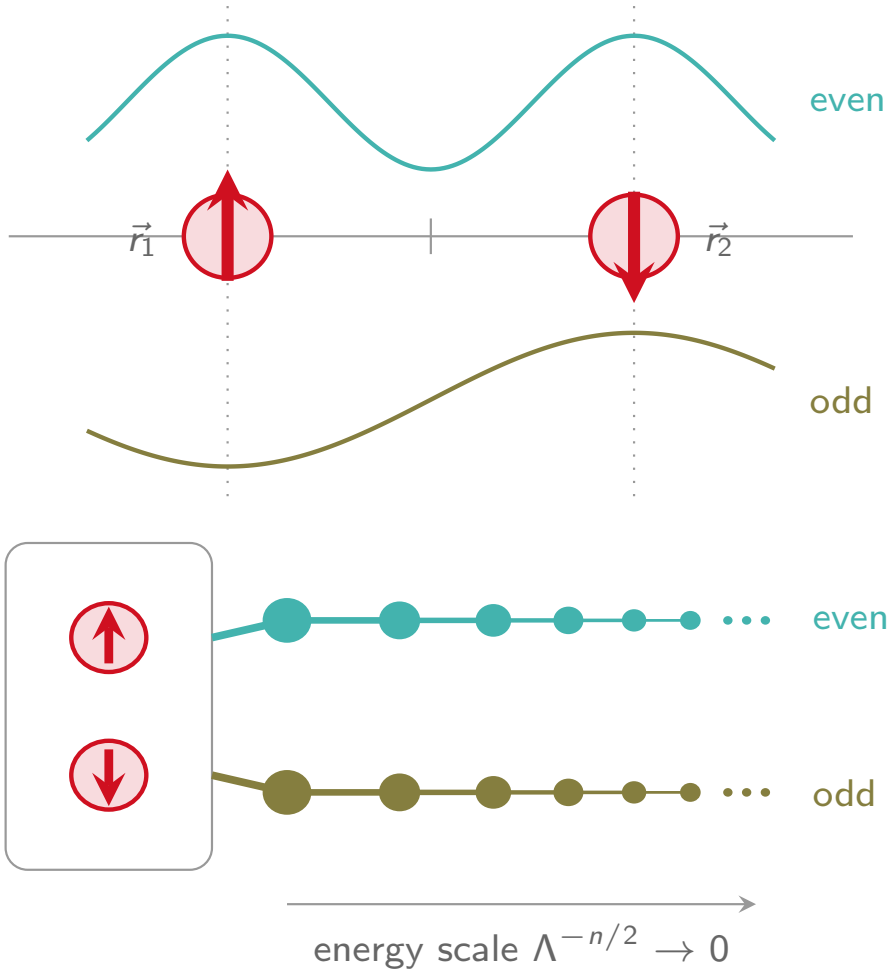
Sort the host electrons by parity about the midpoint

even: $\psi_{\mathrm{e}} \propto \psi(\vec{r}_1) + \psi(\vec{r}_2)$

odd: $\psi_{\mathrm{o}} \propto \psi(\vec{r}_1) - \psi(\vec{r}_2)$

$\psi(\vec{r}_i)$: host electron at impurity site i

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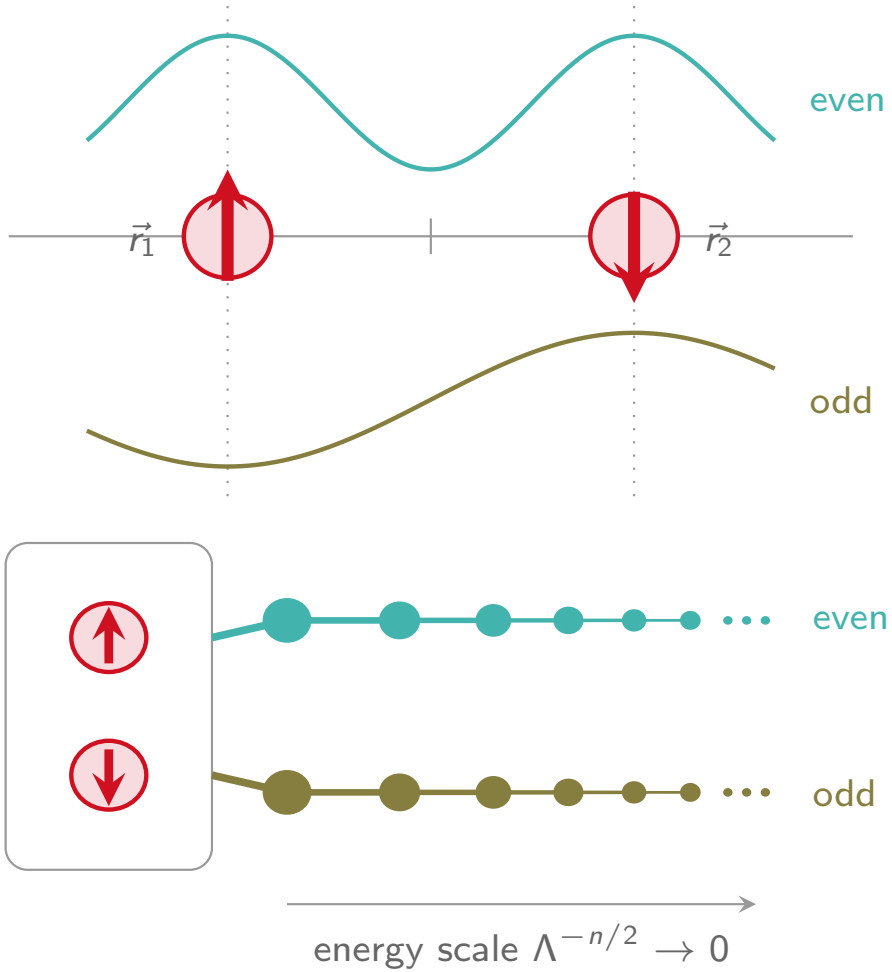
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Each channel: one Wilson chain. NRG treats both spins at once

Jayaprakash, Krishna-murthy, Wilkins, PRL 47, 737 (1981)
Jones, Varma, Wilkins, PRL 61, 125 (1988)
Affleck, Ludwig, PRL 68, 1046 (1992)
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Each channel: one Wilson chain. NRG treats both spins at once

$$\rho_{\rm e/o}(\omega) = \rho(0,\omega) \pm \rho(\vec{r},\omega)$$

local $\pm$ non-local density of states: the only input from the host

Jayaprakash, Krishna-murthy, Wilkins, PRL 47, 737 (1981)
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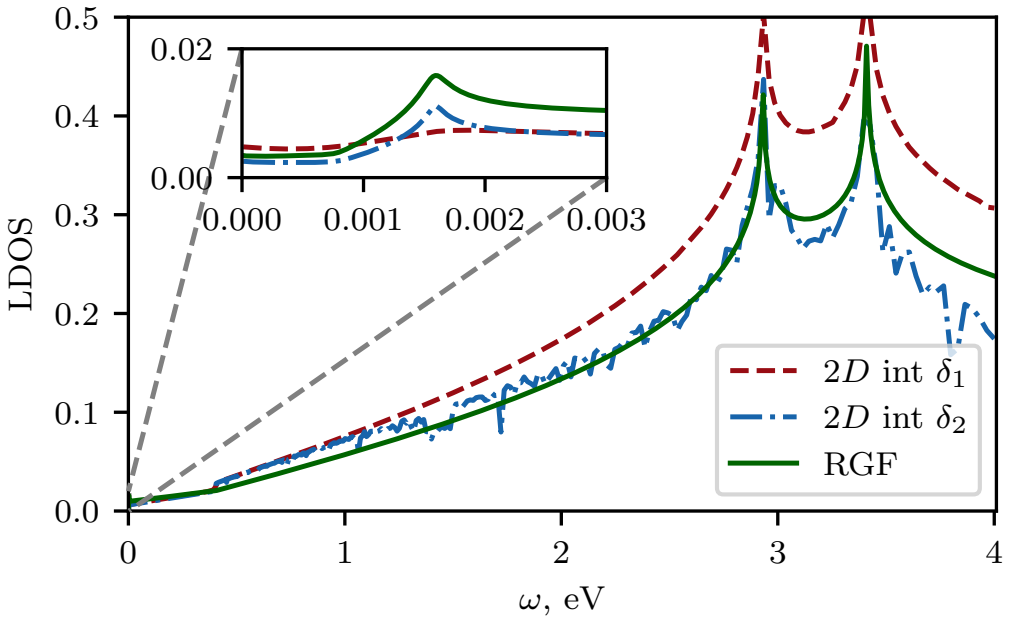
The low-energy local and non-local DoS must be well resolved

The DoS on every energy scale

$$G(\vec{k},\omega) = \left[\omega + i\delta - H(\vec{k})\right]^{-1} \xrightarrow{\text{the impurity is local}} G(\vec{r},\omega) = \int_{\mathrm{BZ}} \frac{d^2k}{(2\pi)^2}\, e^{i\vec{k}\cdot\vec{r}}\, G(\vec{k},\omega)$$

momentum space: immediate from $H(\vec{k})$

real space: a 2D integral at every energy



LDoS on an unstacked site (A1/B2). $\delta_{1,2}$: two broadenings of the 2D integral

NRG logarithmic energy grid, down to far below 1 meV

van Hove peak: $\sim 10^{-4}$ of the Brillouin zone

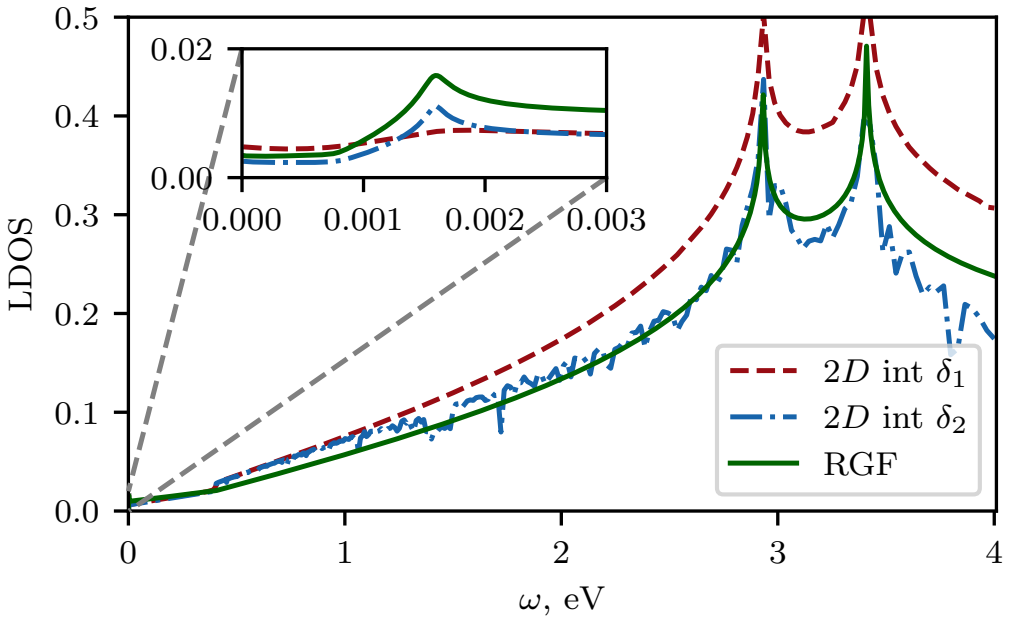
RGF recursive Green's functions: smooth, peak resolved

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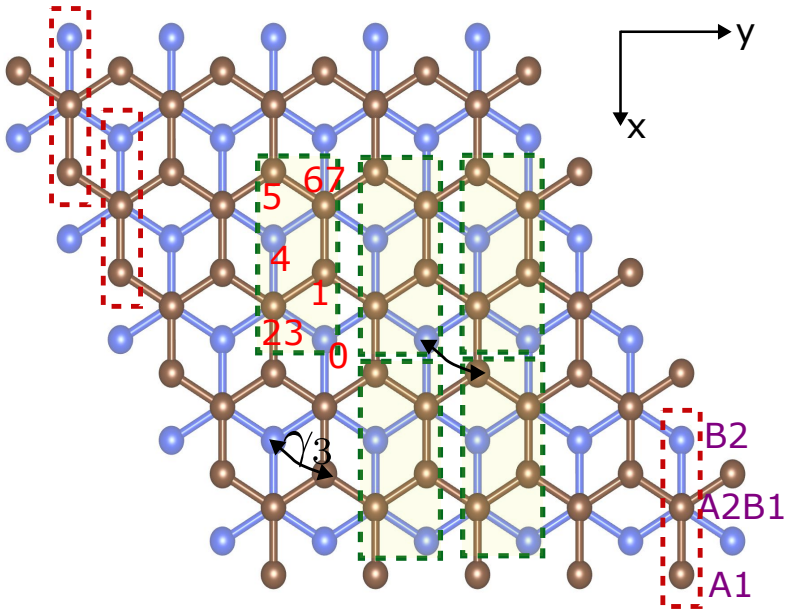
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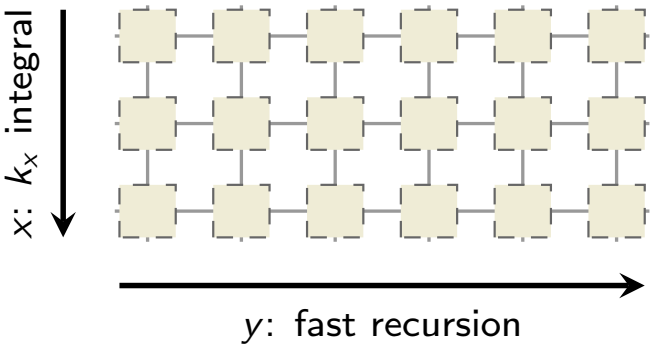
RGF recursive Green's functions: smooth, peak resolved

Direct integration is noisy: unsuitable for the logarithmic grid of NRG

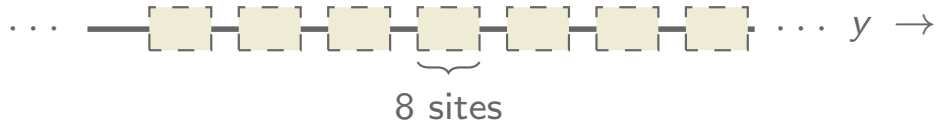
Fast recursive Green's functions



Bilayer graphene redrawn as a square lattice of 8-site cells



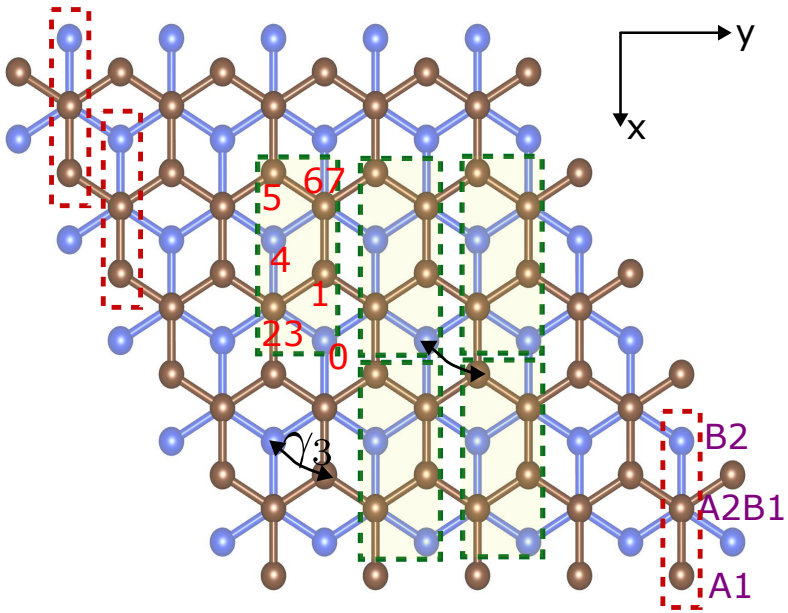
1. Fourier transform along x:
   for each k_x, a 1D chain of cells along y



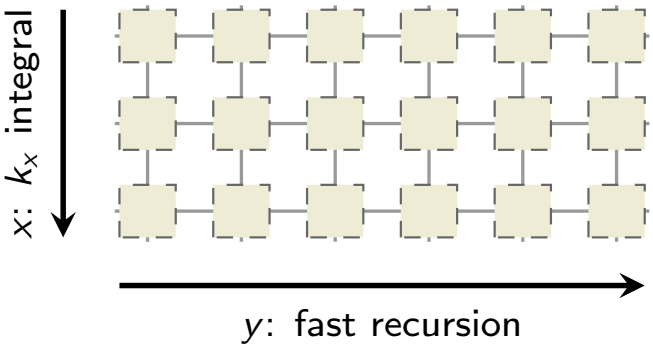
Cuba (Cuhre): Hahn, Comput. Phys. Commun. 168, 78 (2005)
Python interface: Buchner, github.com/JohannesBuchner/cuba

López Sancho, López Sancho, Rubio, J. Phys. F: Met. Phys. 14, 1205 (1984)
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Fast recursive Green's functions

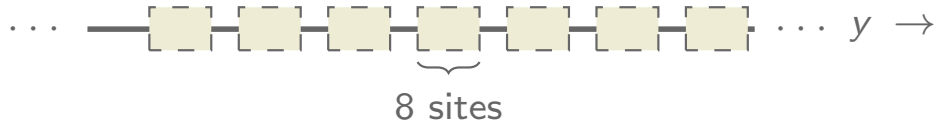


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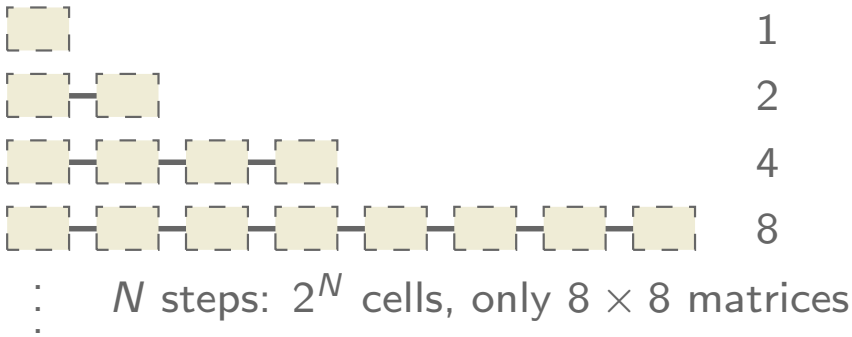


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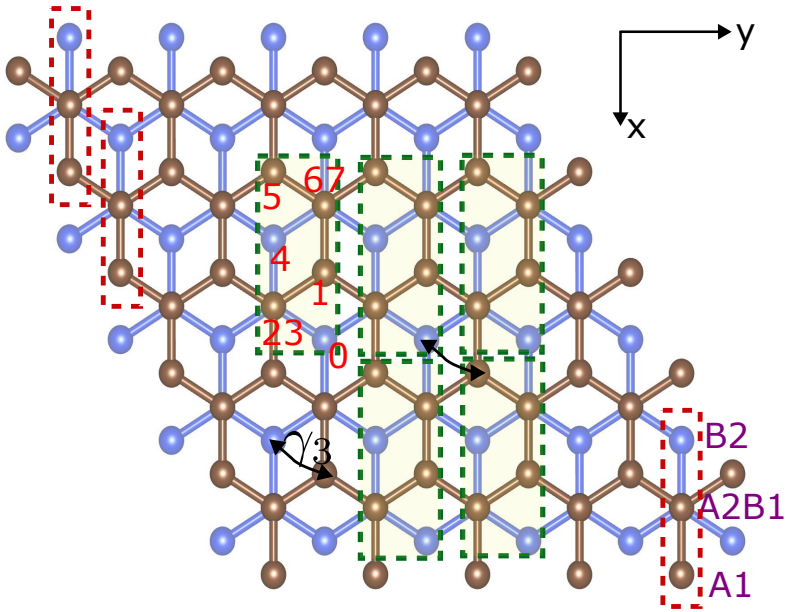


2. Recursion: each step doubles the chain

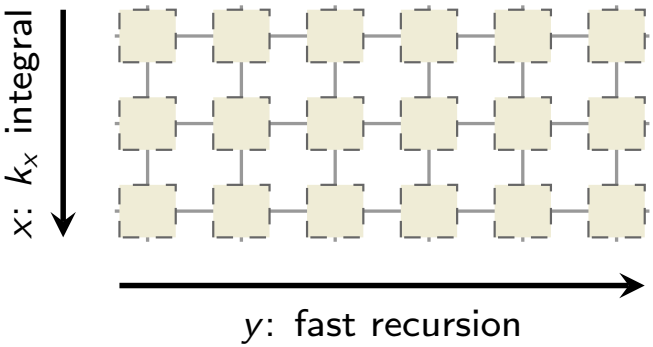


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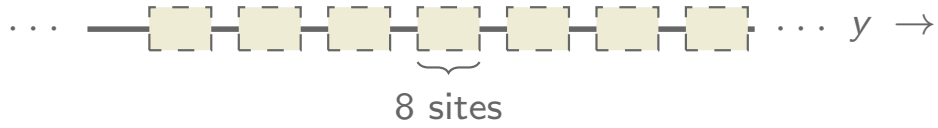


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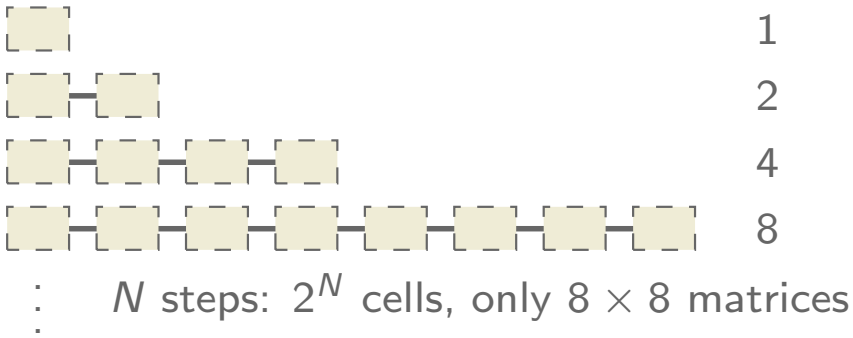


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N steps: 2^N cells, only 8×8 matrices

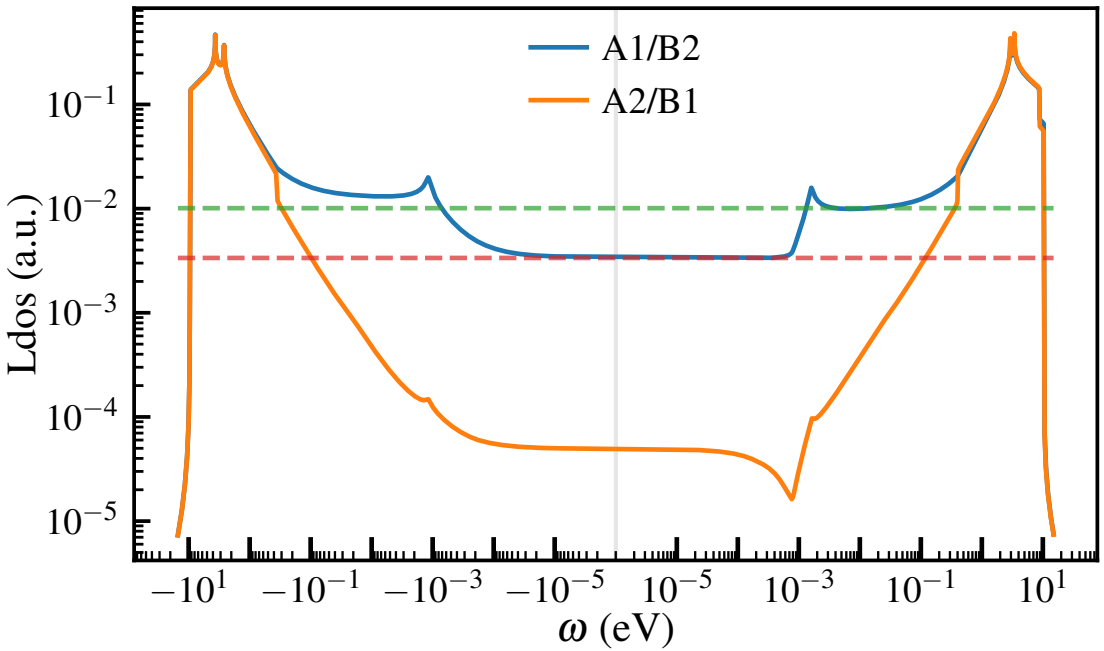
3. Adaptive integral over k_x (Cuhre):

$$G(\vec{r},\omega) = \int \frac{dk_x}{2\pi}\, e^{ik_x x}\, G_{\text{chain}}(k_x,\omega)$$

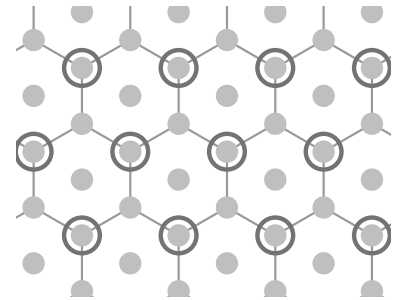
$$\Rightarrow \quad \rho(0,\omega),\ \ \rho(\vec{r},\omega)$$

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Dilute limit: one impurity, two kinds of site



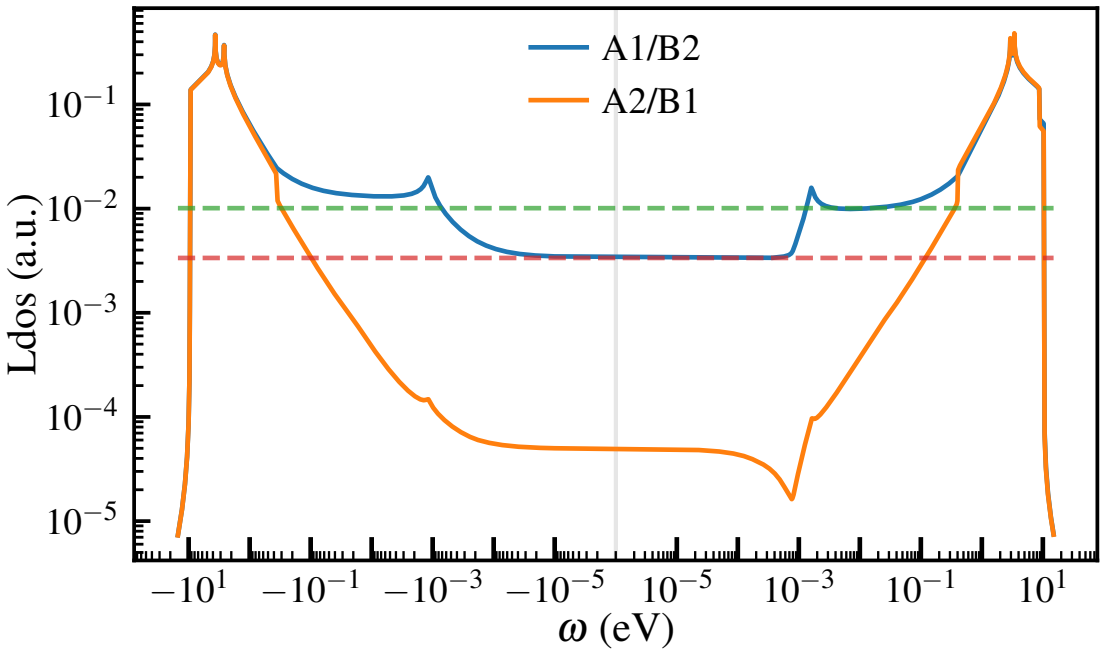
Local density of states from recursive Green's functions



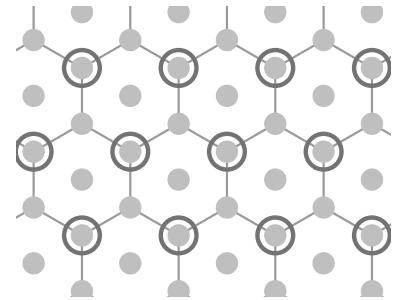
unstacked: A1, B2

stacked: A2/B1

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Local density of states from recursive Green's functions



Unstacked A1/B2

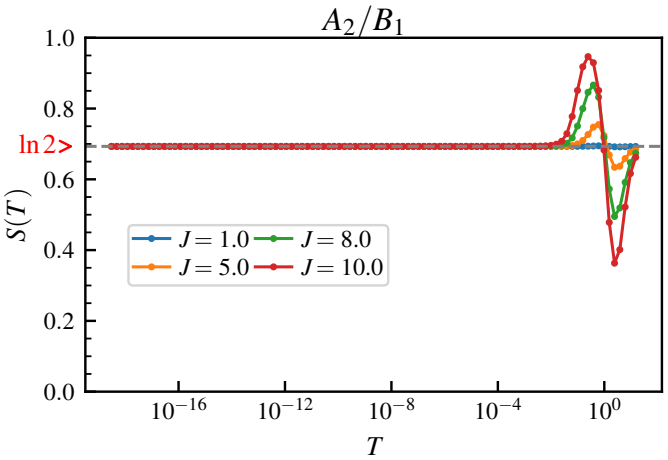
van Hove peak at 1.6 meV,
between two plateaus

Stacked A2/B1

100 times fewer states
at low energy

One impurity: Kondo only on unstacked sites

Stacked A2/B1

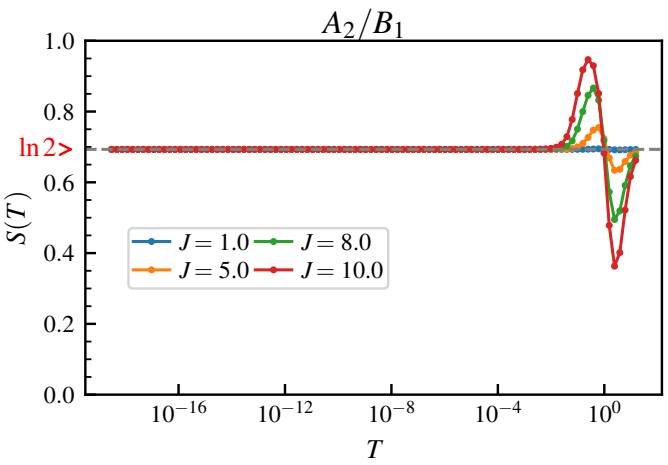


entropy stays at ln 2

free moment

One impurity: Kondo only on unstacked sites

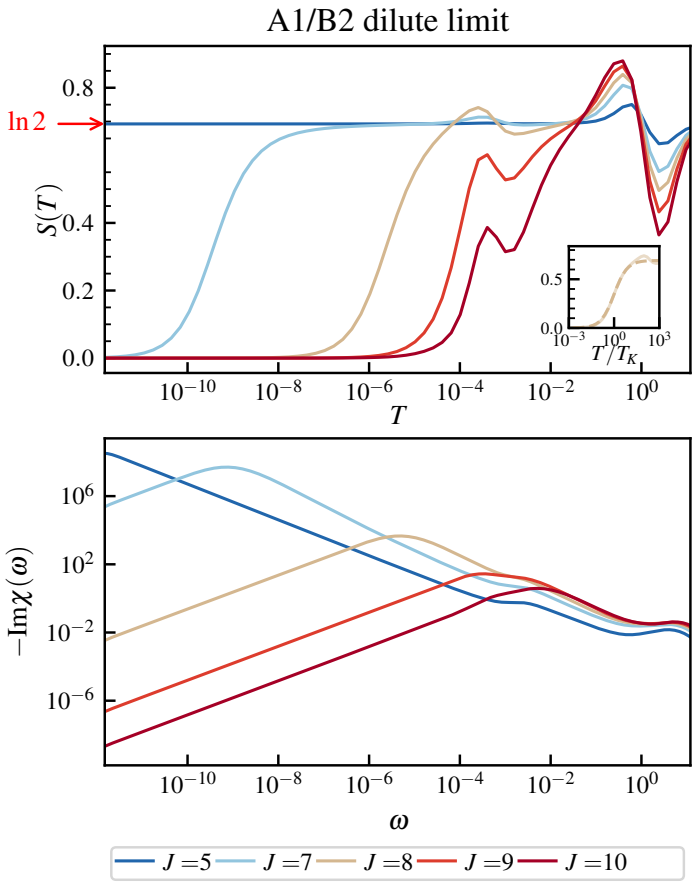
Stacked A2/B1



entropy stays at ln 2

free moment

Unstacked A1/B2



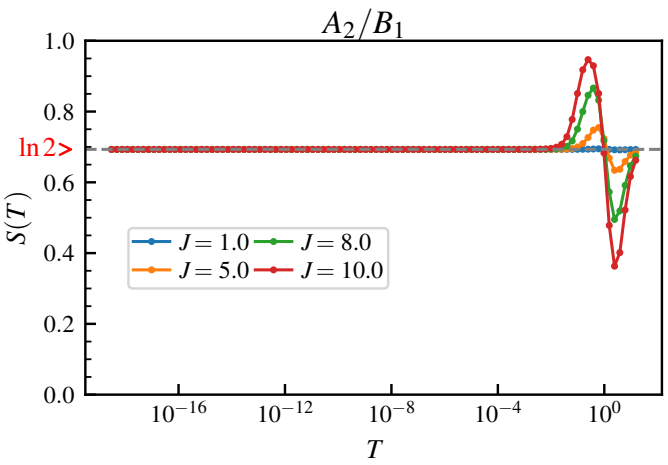
entropy $\ln 2 \to 0$

at strong coupling:

Kondo singlet

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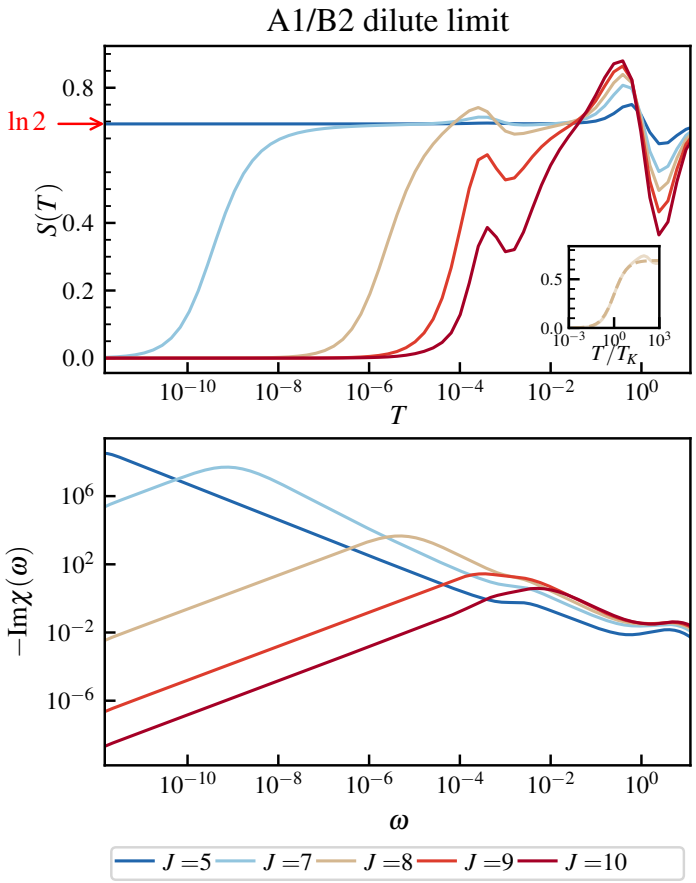
Stacked A2/B1



entropy stays at ln 2

free moment

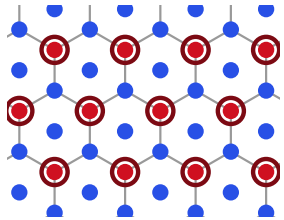
Unstacked A1/B2



entropy ln 2 → 0

at strong coupling:

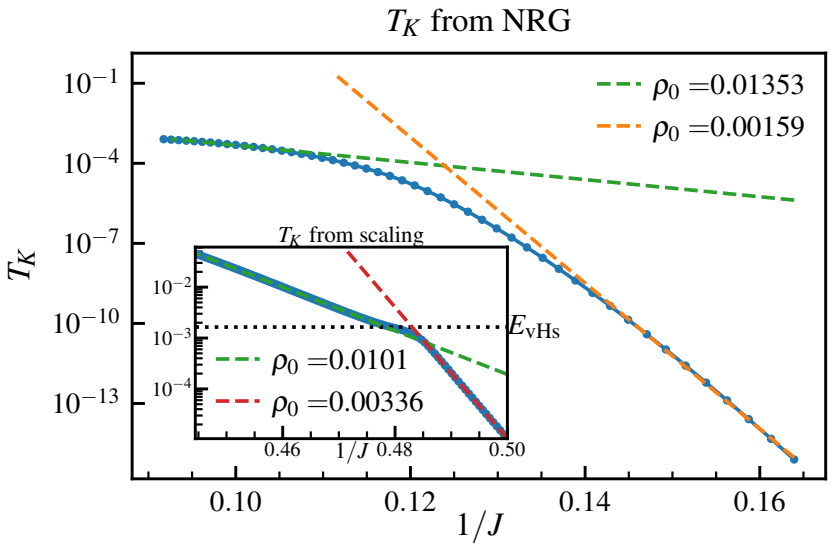
Kondo singlet



- free moment (A2/B1)
- Kondo (A1/B2)

Free moment or Kondo singlet: the site alone decides

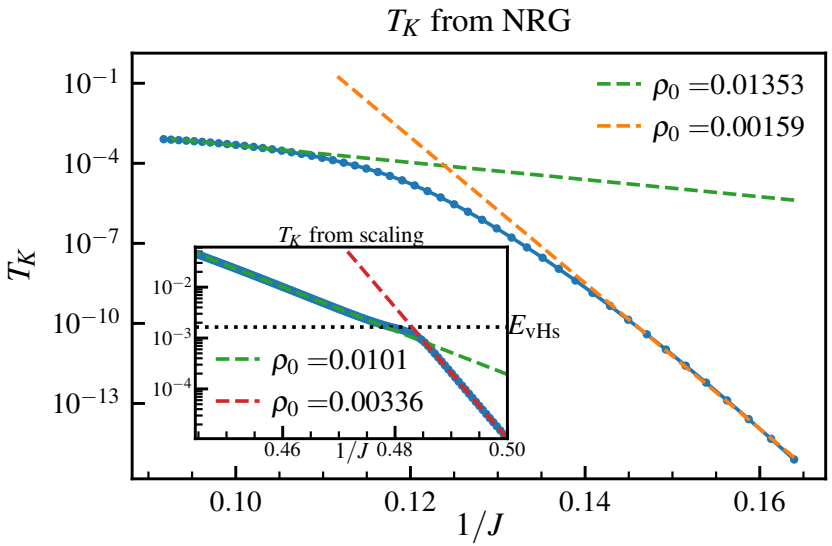
The Kondo scale follows two laws



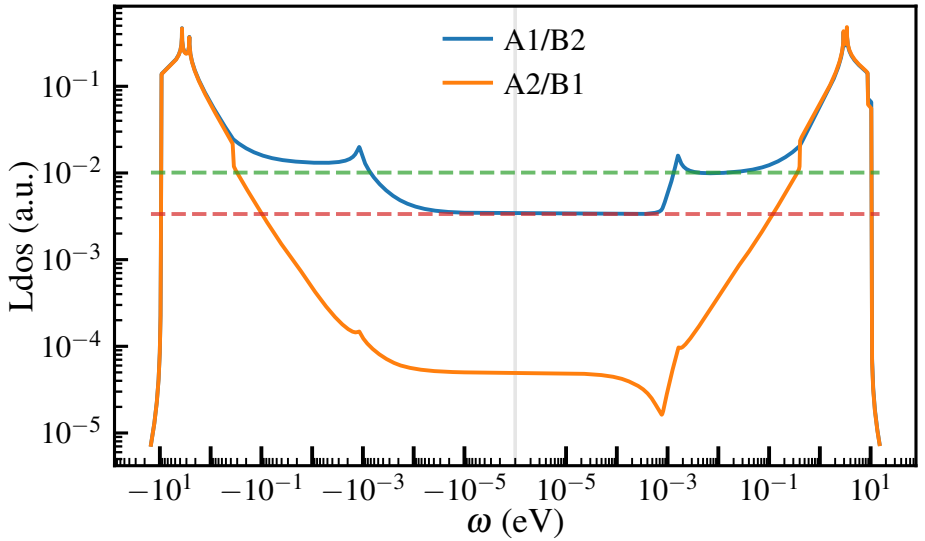
T_K on the unstacked site A1/B2, NRG
Inset: poor man's scaling, Anderson, J. Phys. C 3, 2436 (1970)

$$T_K \sim D\, e^{-1/\rho_{\mathrm{eff}} J}$$

The Kondo scale follows two laws



T_K on the unstacked site A1/B2, NRG
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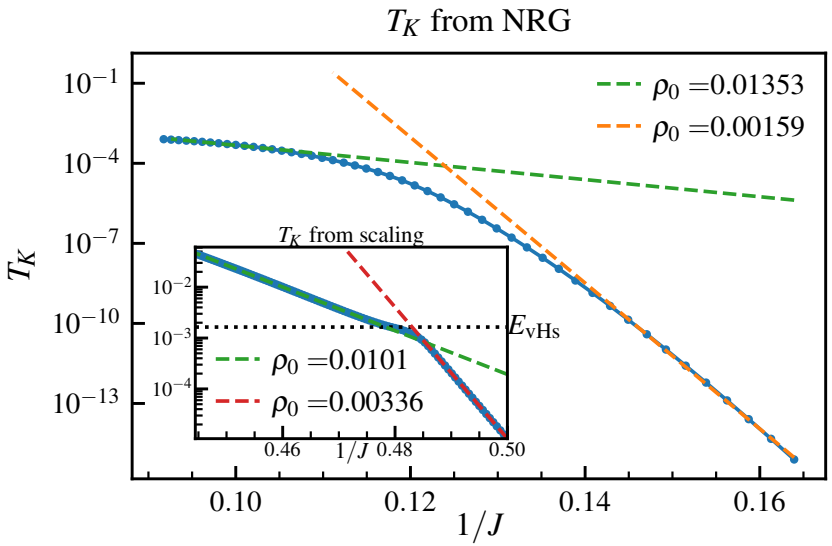
Local density of states
Dashed lines: the two plateaus of A1/B2

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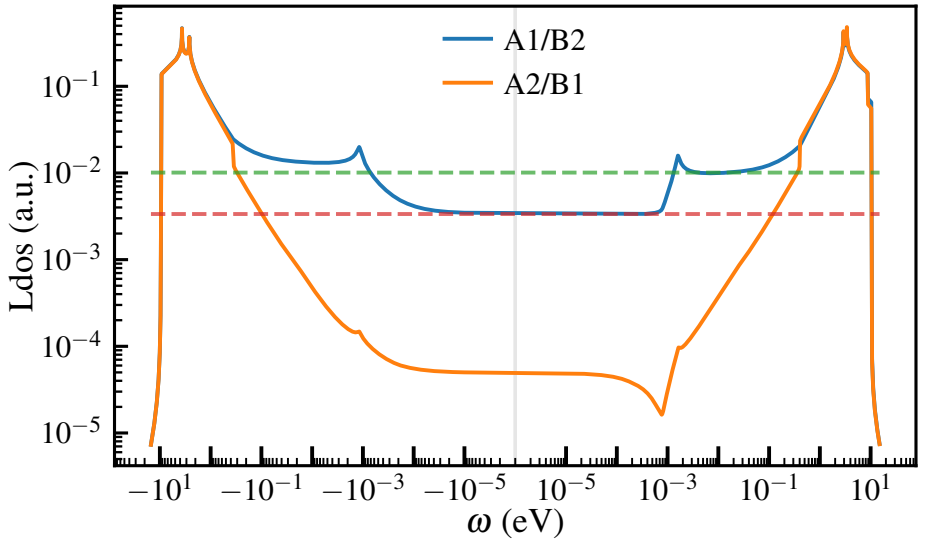
weak J: T_K below the van Hove peak, ρ_{eff} = low plateau

strong J: T_K above the van Hove peak, ρ_{eff} = high plateau

The Kondo scale follows two laws



T_K on the unstacked site A1/B2, NRG
Inset: poor man's scaling, Anderson, J. Phys. C 3, 2436 (1970)



Local density of states
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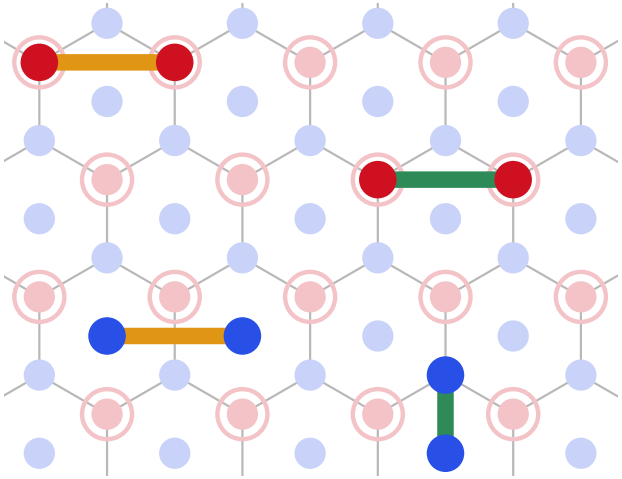
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weak J: T_K below the van Hove peak, ρ_{eff} = low plateau

strong J: T_K above the van Hove peak, ρ_{eff} = high plateau

One site, two Kondo laws: the impurity reads out the van Hove singularity

Two impurities: four kinds of pair



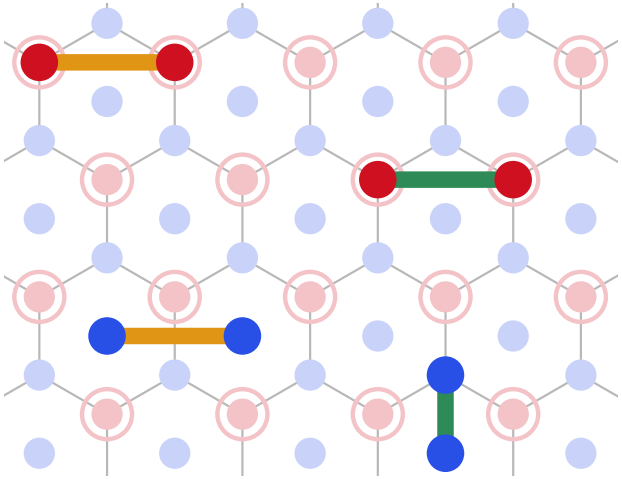
same sublattice: FM

opposite sublattice: AFM

Saremi, PRB 76, 184430 (2007)

pair	RKKY	each alone
A2–A2	FM	free moment
A2–B1	AFM	free moment
A1–B2	AFM	Kondo
B2–B2	FM	Kondo

Two impurities: four kinds of pair



same sublattice: FM

opposite sublattice: AFM

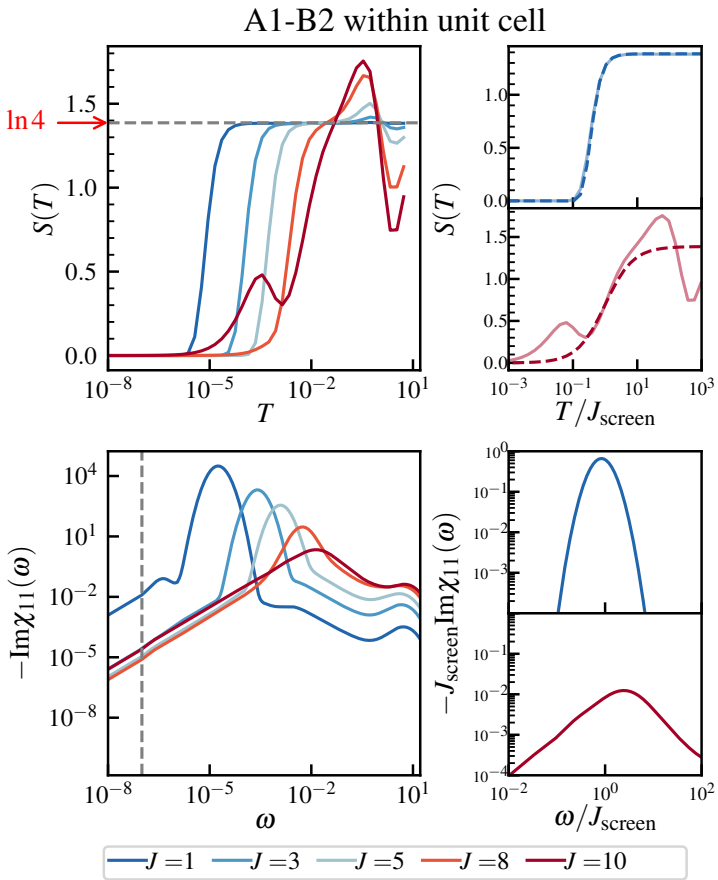
Saremi, PRB 76, 184430 (2007)

pair	RKKY	each alone	
A2–A2	FM	free moment	RKKY only
A2–B1	AFM	free moment	RKKY only
A1–B2	AFM	Kondo	RKKY vs Kondo
B2–B2	FM	Kondo	RKKY vs Kondo

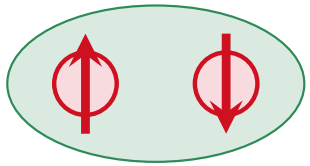
A1–B2: from AFM singlet to Kondo

Kondo model two spins $\frac{1}{2}$, so $\ln 4$ = two free spins

(twisted bilayer part: Anderson model, $\ln 4$ = four orbital states)



Top: entropy. Bottom: spin susceptibility.
Right: rescaled by J_{screen}; dashed: universal curves



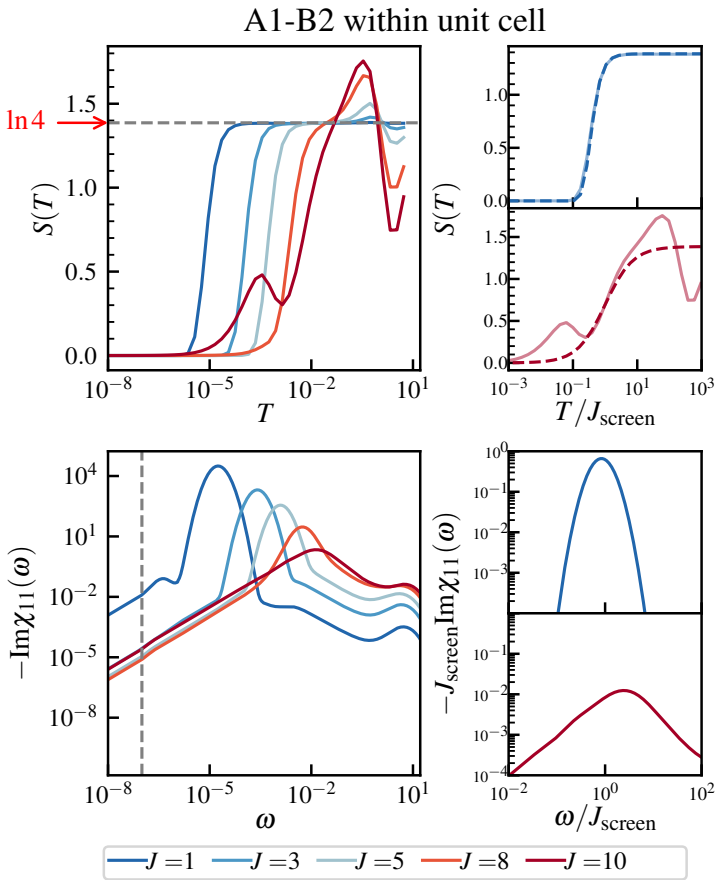
weak J: AFM singlet

the moments bind each other
$\ln 4 \to 0$ in one step
sharp peak at J_{RKKY}

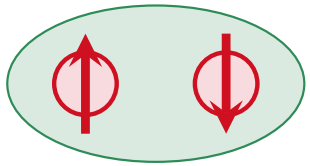
A1–B2: from AFM singlet to Kondo

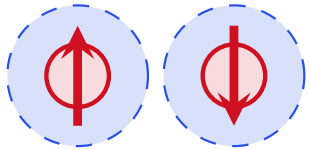
Kondo model two spins $\frac{1}{2}$, so $\ln 4$ = two free spins

(twisted bilayer part: Anderson model, $\ln 4$ = four orbital states)



Top: entropy. Bottom: spin susceptibility.
Right: rescaled by J_{screen}; dashed: universal curves





weak J: AFM singlet

the moments bind each other
$\ln 4 \to 0$ in one step
sharp peak at J_{RKKY}

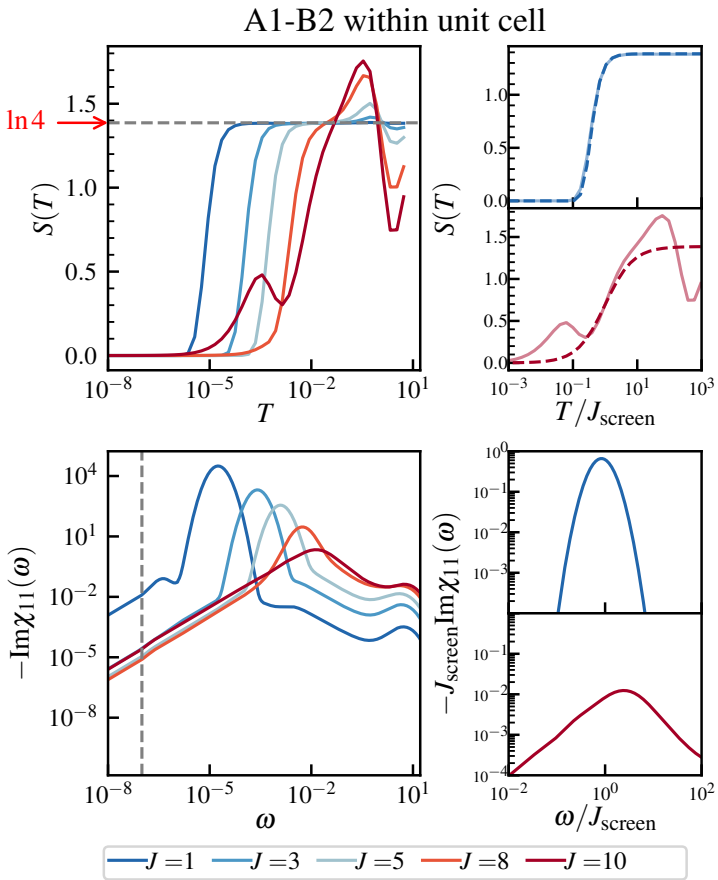
strong J: Kondo

each moment screened alone
broad crossover to 0
broad Kondo peak

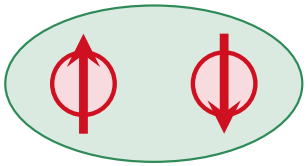
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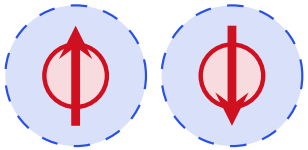
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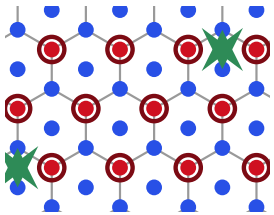
(twisted bilayer part: Anderson model, $\ln 4$ = four orbital states)



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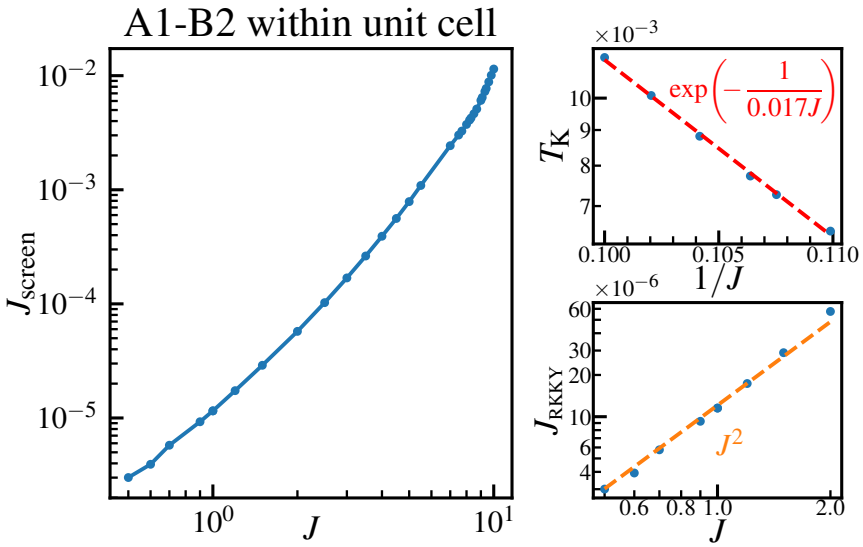
strong J: Kondo

each moment screened alone
broad crossover to 0
broad Kondo peak

- free moment (A2/B1)
- Kondo (A1/B2)
- AFM singlet (A1–B2)

A1–B2: one screening scale, two regimes

against the coupling J



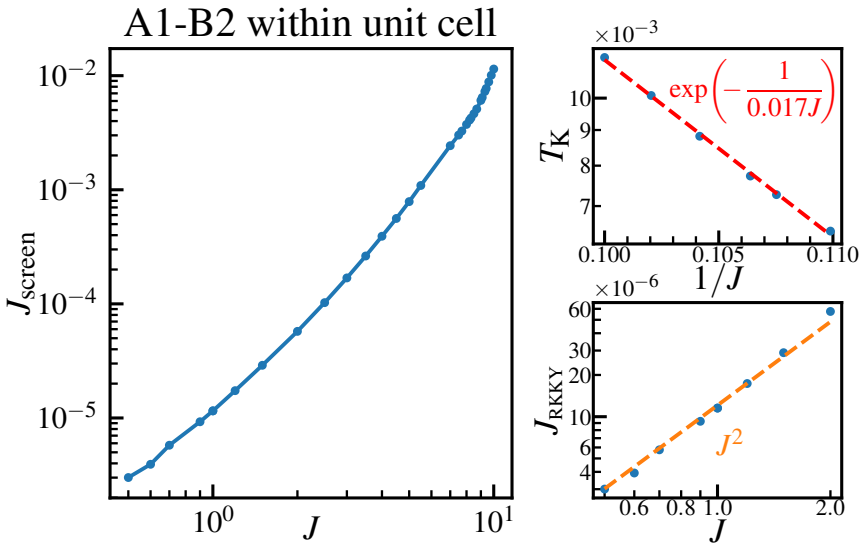
$J_{\rm screen}$ from the entropy, $S(J_{\rm screen}) = 1$

weak J: $J_{\rm screen} \propto J^2$

strong J: $J_{\rm screen} \sim e^{-1/0.017J}$

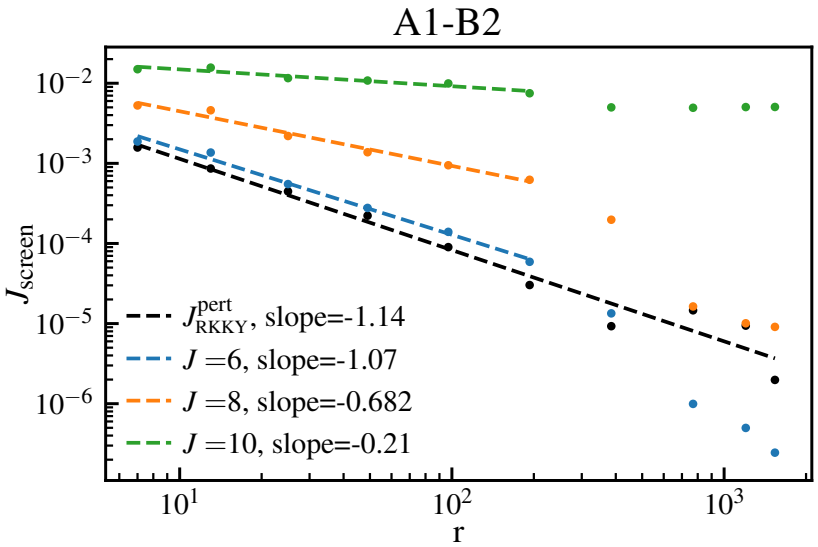
A1–B2: one screening scale, two regimes

against the coupling J



$J_{\rm screen}$ from the entropy, $S(J_{\rm screen}) = 1$

against the distance r



$J_{\rm screen}$ from the peak of the spin susceptibility

weak J: $J_{\rm screen} \propto J^2$

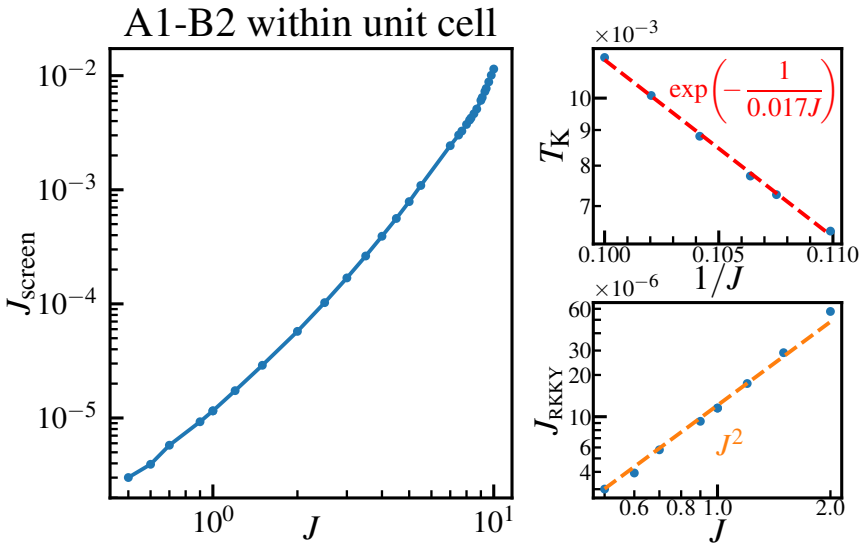
strong J: $J_{\rm screen} \sim e^{-1/0.017J}$

weak J: $J_{\rm screen} \sim r^{-1.14}$

$J = 10$: $J_{\rm screen} \sim r^{-0.21}$, nearly flat

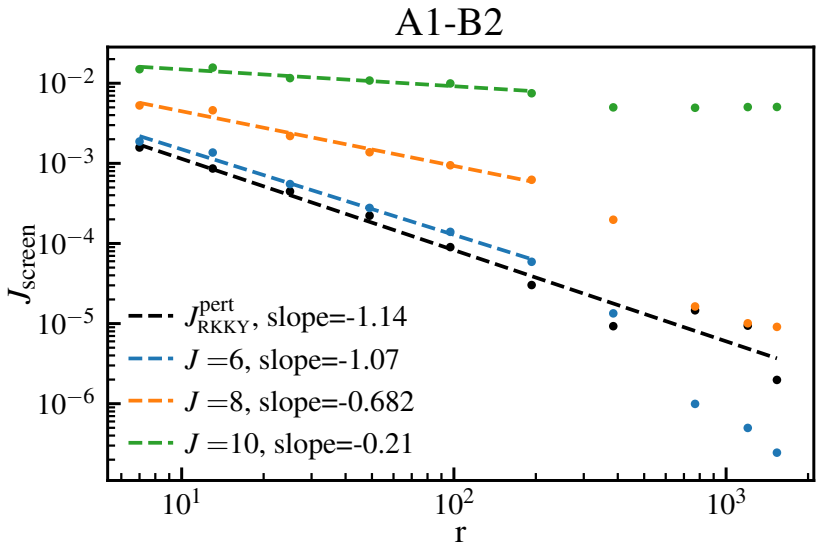
A1–B2: one screening scale, two regimes

against the coupling J



J_{screen} from the entropy, $S(J_{\mathrm{screen}}) = 1$

against the distance r



J_{screen} from the peak of the spin susceptibility

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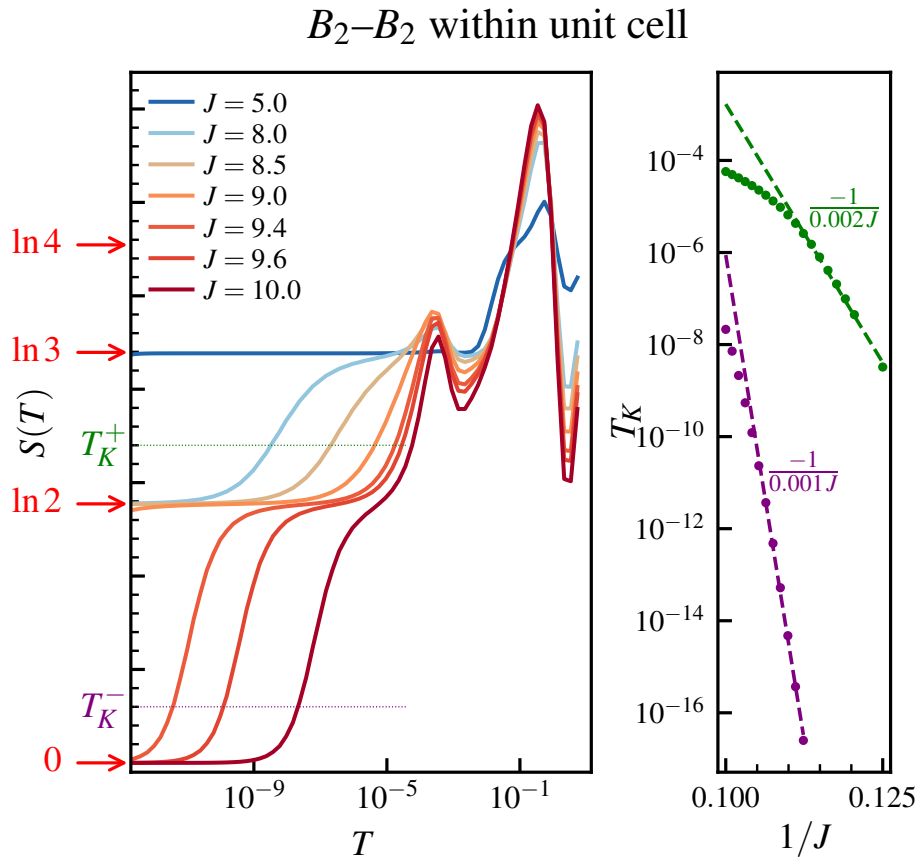
Weak J: the pair acts together. Strong J: each impurity on its own

B2–B2: ferromagnetic lock, then two-stage Kondo

two free spins

FM lock: spin 1





Two impurities on B2 sites of one unit cell. Right: both $T_K^{\pm}$ exponential in $1/J$

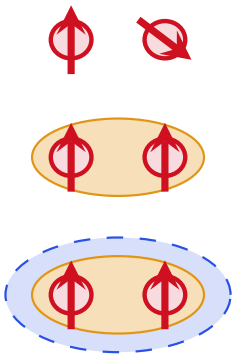
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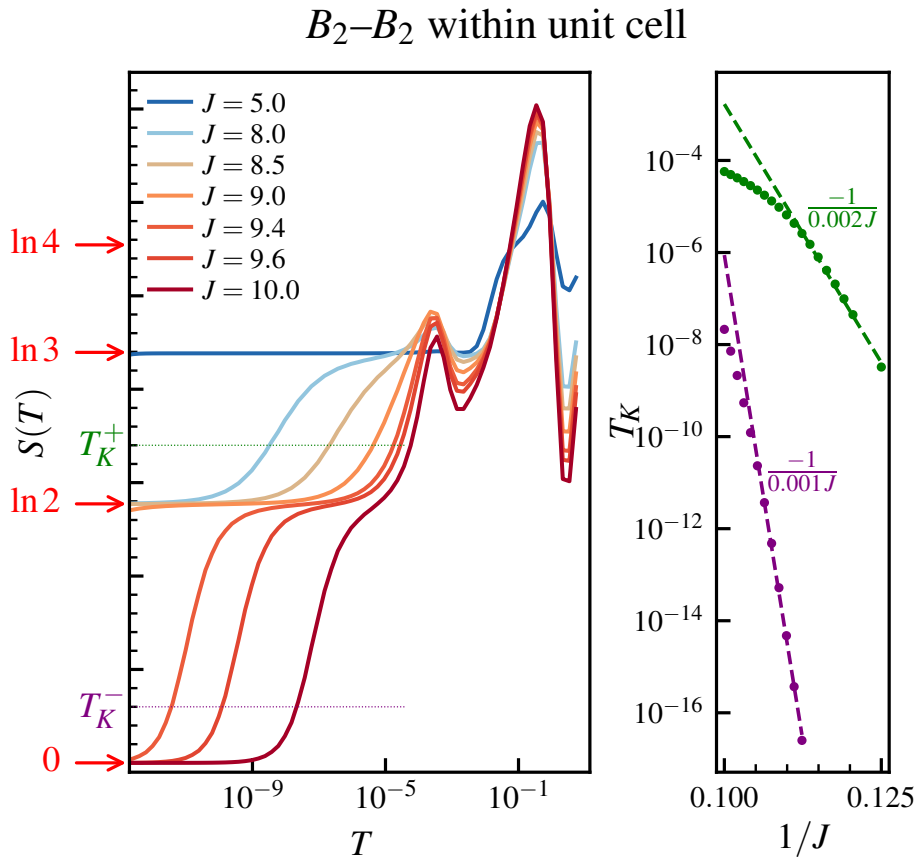
two free spins

FM lock: spin 1

even channel screens

below T_K^+: spin $\frac{1}{2}$ left





Two impurities on B2 sites of one unit cell. Right: both $T_K^{\pm}$ exponential in $1/J$

B2–B2: ferromagnetic lock, then two-stage Kondo

two free spins

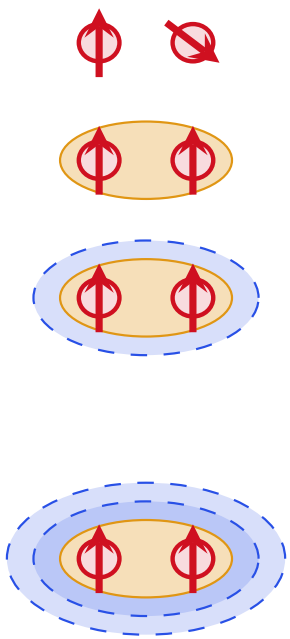
FM lock: spin 1

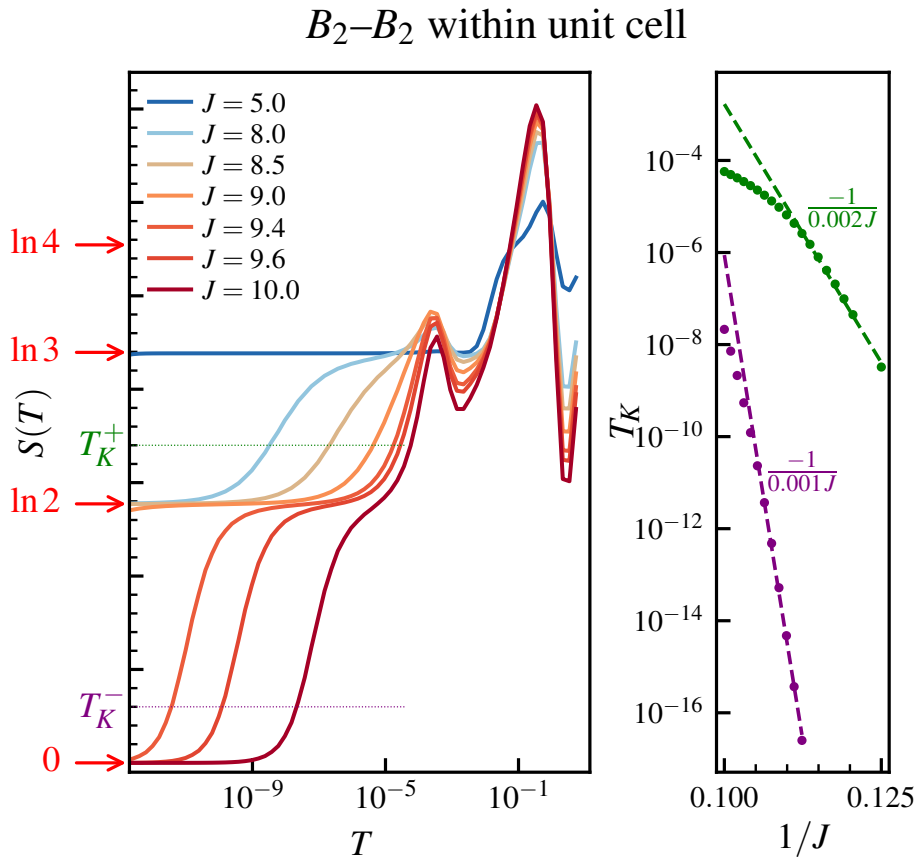
even channel screens

below T_K^+: spin $\frac{1}{2}$ left

odd channel screens

below T_K^-: singlet

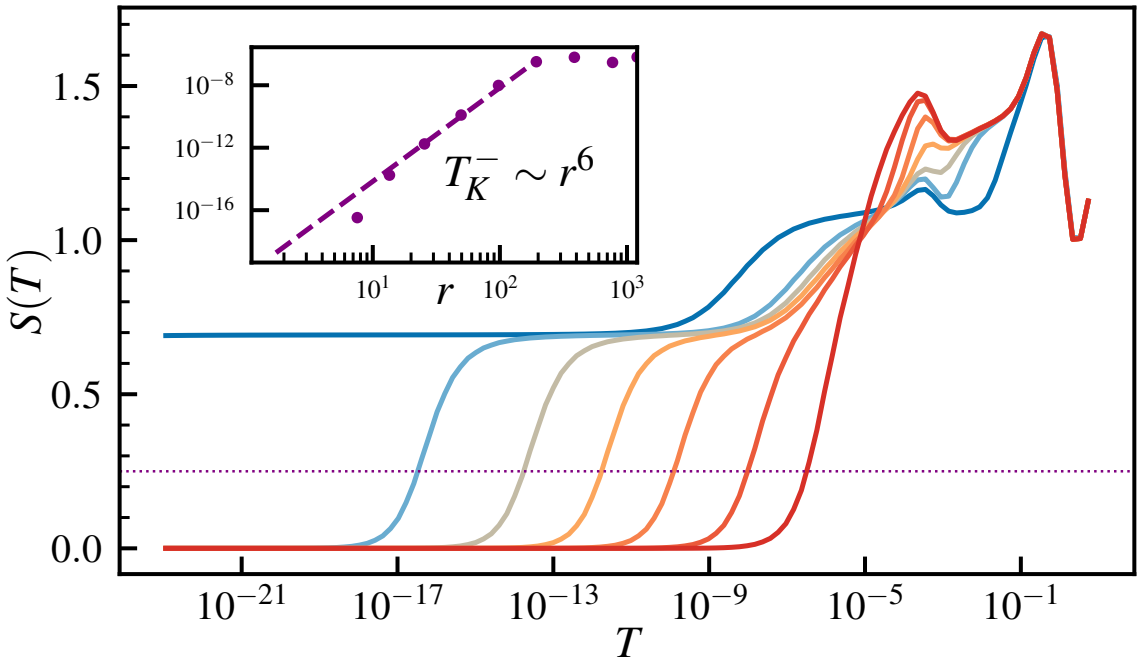




Two impurities on B2 sites of one unit cell. Right: both $T_K^{\pm}$ exponential in $1/J$

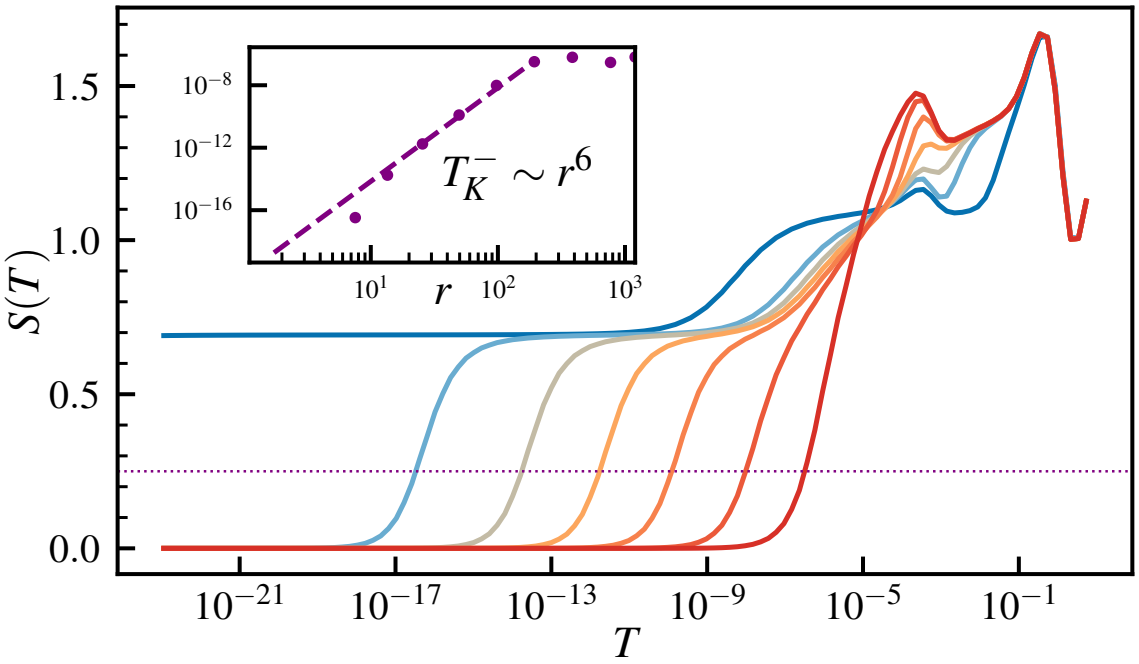
Ferromagnetic RKKY does not block Kondo screening: it splits it into two stages

Further apart: the two stages merge



B2–B2 pairs, x unit cells apart. Inset: T_K^- against the distance r

Further apart: the two stages merge

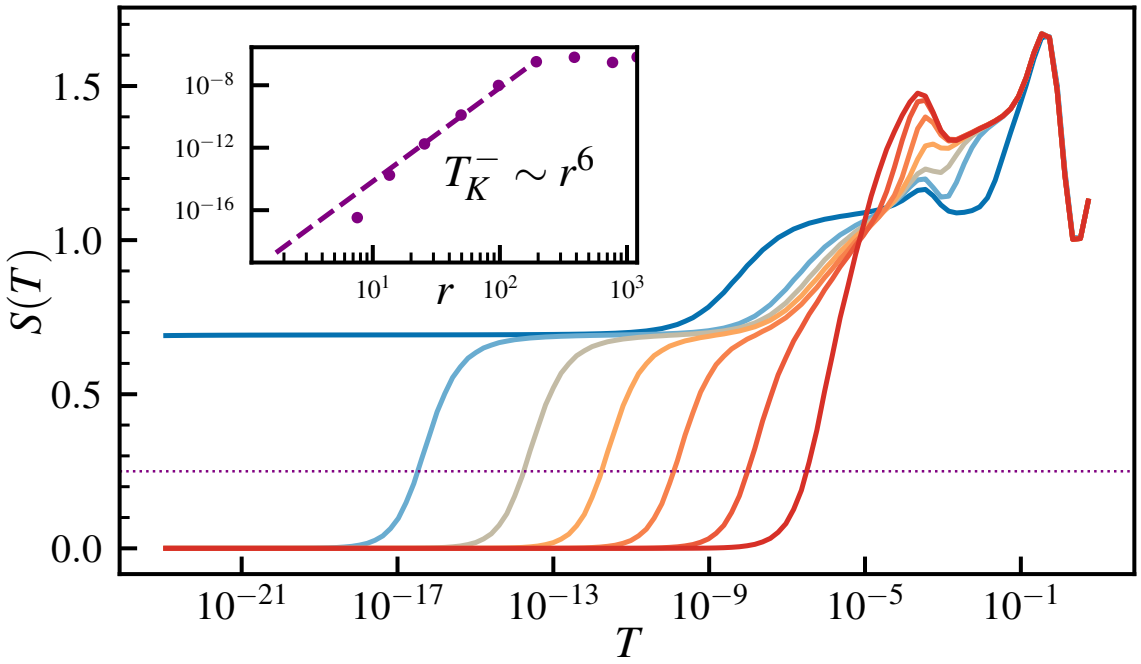


B2–B2 pairs, x unit cells apart. Inset: T_K^- against the distance r

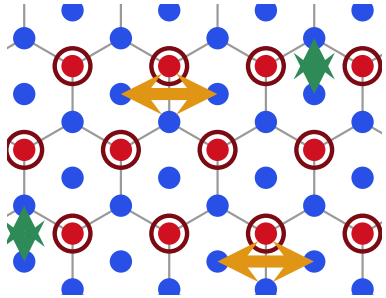
close pair: wide $\ln 2$ plateau, T_K^- far below T_K^+

far apart: $T_K^- \sim r^6$ catches up: one Kondo scale

Further apart: the two stages merge



B2–B2 pairs, x unit cells apart. Inset: T_K^- against the distance r

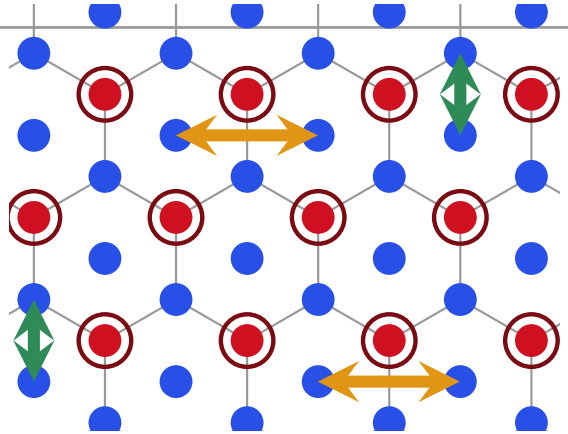


- free moment (A2/B1)
- Kondo (A1/B2)
- AFM singlet (A1–B2)
- FM two-stage (B2–B2)

close pair: wide ln 2 plateau, T_K^- far below T_K^+

far apart: $T_K^- \sim r^6$ catches up: one Kondo scale

One host, every screening regime: the impurity position selects it



- free moment (A2/B1)
- Kondo (A1/B2)
- AFM singlet (A1–B2)
- FM two-stage (B2–B2)

Stacked: free moment, never screened

Unstacked: Kondo, AFM singlet, FM two-stage

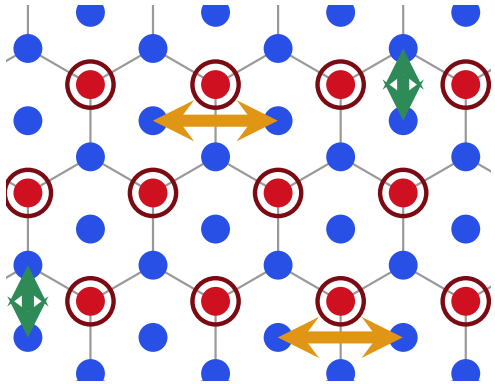
Position and pairing pick the regime

Outlook: STM on BLG, artificial impurity arrays, RKKY in TBG

Site selectivity: every Kondo regime in one sample

Thank you

One host, four screening regimes



- free moment (A2/B1)
- Kondo (A1/B2)
- AFM singlet (A1–B2)
- FM two-stage (B2–B2)

With

D. O. Oriekhov — TU Delft
L. Fritz — Utrecht University
A. K. Mitchell — University College Dublin

Support

Kavli Foundation
DelftBlue supercomputer, TU Delft

The Abdus Salam International Centre for Theoretical Physics

Twisted bilayer: Shankar, Oriekhov, Mitchell, Fritz, PRB 107, 245102 (2023)
Bernal bilayer: Shankar, Oriekhov, Fritz, Mitchell, manuscript in preparation

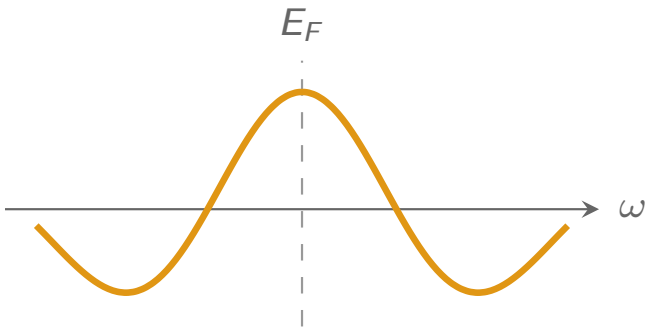
Backup: why the RKKY sign follows the sublattice

Split the non-local DoS by parity in energy:

$$\rho(\vec{r},\omega) = \rho_s(\vec{r},\omega) + \rho_a(\vec{r},\omega), \qquad \rho_{s/a}(-\omega) = \pm\,\rho_{s/a}(\omega)$$

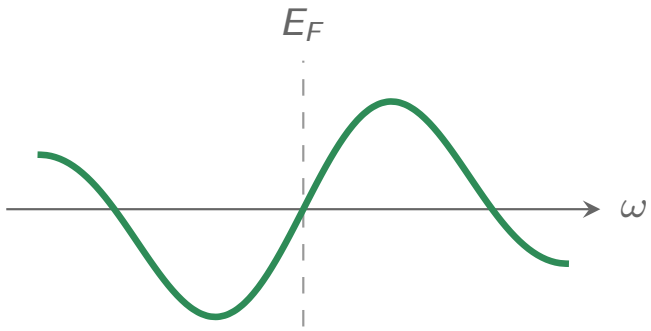
$$J_{\mathrm{RKKY}} = -2J^2 \int_0^W \frac{d\Omega_1}{\pi} \int_0^W \frac{d\Omega_2}{\pi} \, \frac{\rho_s(\Omega_1)\,\rho_s(\Omega_2) \;-\; \rho_a(\Omega_1)\,\rho_a(\Omega_2)}{\Omega_1 + \Omega_2}$$

one loop, particle-hole symmetric band; energies measured from E_F; the denominator is positive



same sublattice

ρ_s dominates: $J_{\mathrm{RKKY}} < 0$, ferromagnetic

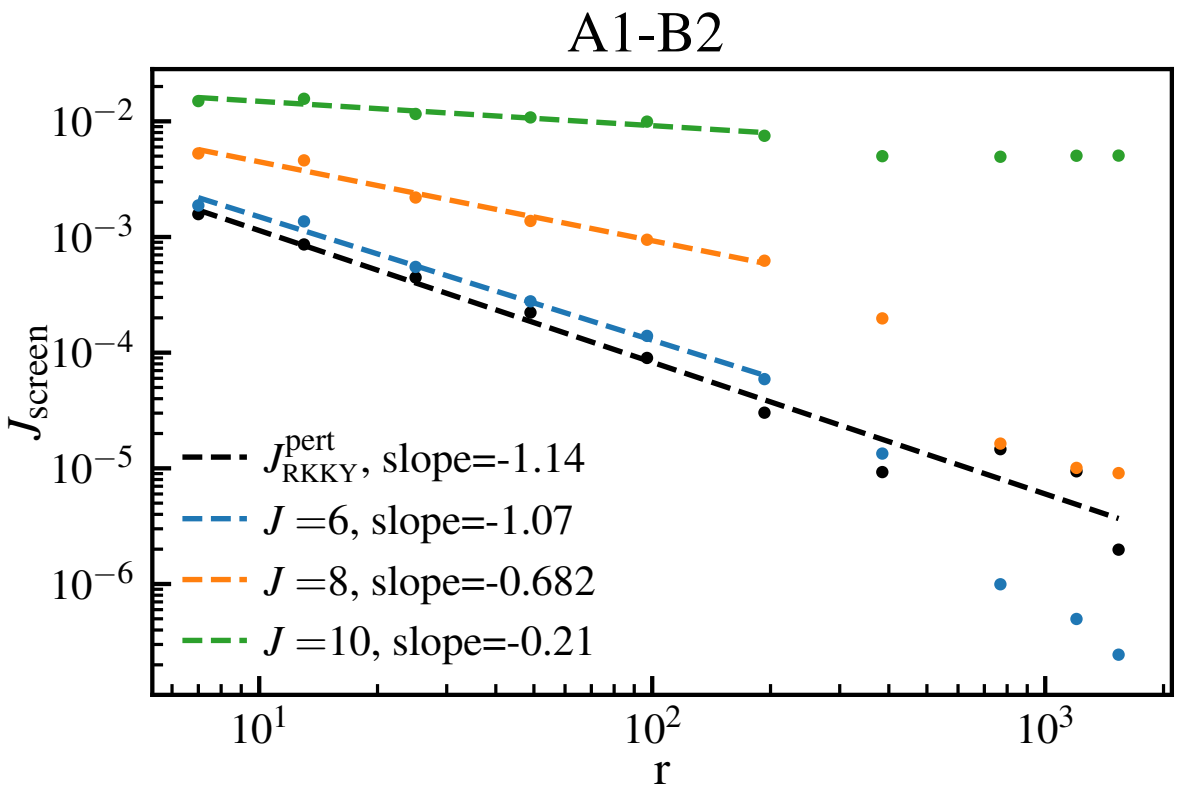


opposite sublattice

ρ_a dominates: $J_{\mathrm{RKKY}} > 0$, antiferromagnetic

Curves: sketches of $\rho(\vec{r},\,\omega)$, even and odd in ω. Saremi, PRB 76, 184430 (2007)

Backup: A1–B2, screening scale against distance



J_{screen} from the peak of the spin susceptibility

coupling	slope
perturbative	-1.14
$J = 6$	-1.07
$J = 8$	-0.68
$J = 10$	-0.21

Intermediate J: RKKY at short distance, single-impurity Kondo further out

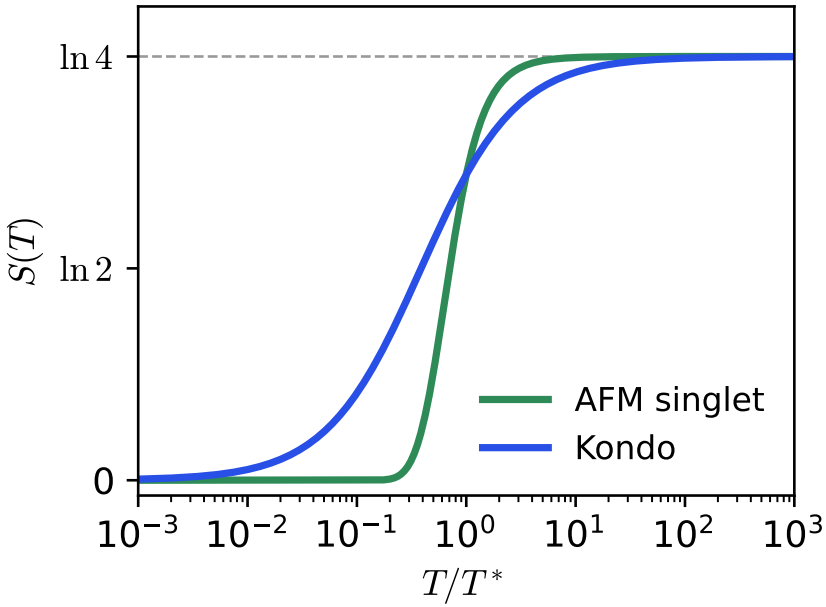
Backup: AFM singlet or Kondo? The shape of the entropy

$$H = J_{\mathrm{RKKY}}\, \vec{S}_1 \cdot \vec{S}_2, \qquad u = e^{|J_{\mathrm{RKKY}}|/T}$$

AFM $\quad S = \ln(u+3) - \dfrac{u \ln u}{u+3}$

Kondo $\quad S \simeq 2 \times \dfrac{\ln 2}{1 + T_K/T}$

FM $\quad S = \ln(1+3u) - \dfrac{3u \ln u}{1+3u} \;\to\; \ln 3$



Both from ln 4 to 0; T^* defined by $S(T^*) = 1$

AFM singlet : a gap, $S \sim e^{-J_{\mathrm{RKKY}}/T}$, gone within one decade

Kondo : no gap, $S \sim T/T_K$, spread over several decades

Backup: toy model, RKKY develops a logarithmic decay

